Emergent Mathematics

Paul Gustafson

Contents

Part I

Symmetry

1 Spectral Theory and Quantum Spectroscopy

1.1 Linear spaces, tensor products, and duality

Definition 1.1 (Linear spaces and maps)**.** Let k be $\mathbb{R}$ or $\mathbb{C}$. A *linear space* over k is an abelian group V with scalar multiplication satisfying

$$a(v+w) = av + aw, \quad (a+b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map $T : V \to W$ is *linear* when $T(av + bw) = aT(v) + bT(w)$. Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For $T : V \to V$, a scalar λ is an *eigenvalue* when $\ker(T - \lambda I) \neq 0$; a nonzero vector in this kernel is an *eigenvector*.

Definition 1.2 (Dual spaces and tensor products)**.** The *dual space* of V is $V^\vee = \mathrm{Hom}_k(V, k)$. Its canonical evaluation pairing is

$$\mathrm{ev}_V : V^\vee \times V \longrightarrow k, \qquad (\varphi, v) \longmapsto \varphi(v).$$

The algebraic transpose of $T : V \to W$ is $T^\vee : W^\vee \to V^\vee$, defined by $T^\vee(\varphi) = \varphi \circ T$.

The *tensor product* $V \otimes W$ is characterized by a bilinear map $(v, w) \mapsto v \otimes w$ with the following universal property: every bilinear map $b : V \times W \to X$ factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\widetilde{b}} X.$$

The evaluation pairing induces a linear map $\mathrm{ev}_V : V^\vee \otimes V \to k$.

Definition 1.3 (Finite-dimensional linear spaces)**.** A linear space V is *finite-dimensional* when there is a coevaluation map

$$\mathrm{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^\vee}) \circ (\mathrm{id}_{V^\vee} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^\vee}.$$

Definition 1.4 (Trace and dimension)**.** For a finite-dimensional linear space V, let $s_{V,V^\vee} : V \otimes V^\vee \to V^\vee \otimes V$ exchange the factors. For $T \in \mathrm{End}(V)$, define

$$\mathrm{tr}(T) = \mathrm{ev}_V \circ s_{V,V^\vee} \circ (T \otimes \mathrm{id}_{V^\vee}) \circ \mathrm{coev}_V, \qquad \dim V = \mathrm{tr}(\mathrm{id}_V).$$

1.2 Exterior and symmetric powers

Definition 1.5 (Exterior powers and determinant)**.** For $m \geq 0$, the mth *exterior power* of a linear space V is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of $v_1 \otimes \cdots \otimes v_m$ is written $v_1 \wedge \cdots \wedge v_m$. An endomorphism T induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = Tv_1 \wedge \cdots \wedge Tv_m.$$

If V has finite dimension n, its *determinant* is the scalar $\det(T)$ defined by the action on the highest nonzero exterior power:

$$\Lambda^n T = \det(T) \operatorname{id}_{\Lambda^n V}.$$

Exercise 1.1 (Exterior powers and determinant)**.** For a finite-dimensional linear space V of dimension n, prove

$$\dim \Lambda^m V = \binom{n}{m}, \qquad \Lambda^m V = 0 \quad (m > n).$$

Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S)\det(T), \qquad T \text{ is invertible} \iff \det(T) \neq 0.$$

Definition 1.6 (Symmetric powers)**.** For a linear space V, the mth *symmetric power* is

$$\operatorname{Sym}^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m - v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(m)} : \sigma \in S_m \rangle.$$

An endomorphism T induces $\operatorname{Sym}^m T$ on $\operatorname{Sym}^m V$.

Exercise 1.2 (Symmetric-power trace identity)**.** Let V be a finite-dimensional complex linear space and $T \in \operatorname{End}(V)$. Prove that

$$\sum_{m=0}^{\infty} \operatorname{tr}(\operatorname{Sym}^m T) z^m = \frac{1}{\det(I - zT)}$$

for $|z|$ smaller than the reciprocal of the spectral radius of T. For $T = I$, deduce

$$\sum_{m=0}^{\infty} \dim(\operatorname{Sym}^m V) z^m = (1 - z)^{-\dim V}.$$

1.3 Hilbert spaces and trace ideals

Definition 1.7 (Hilbert spaces and orthogonality)**.** An *inner product* on a complex linear space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies $\langle x, y \rangle = \overline{\langle y, x \rangle}$. Its norm is $\|x\| = \sqrt{\langle x, x \rangle}$. A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace $M \subseteq H$, define

$$M^{\perp} = \{x : \langle m, x \rangle = 0 \text{ for every } m \in M\}.$$

Definition 1.8 (Bounded operators and adjoints)**.** A linear map $T : H \to K$ between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\|\le 1} \|Tx\| < \infty.$$

An *adjoint* of T is an operator $T^* : K \to H$ satisfying

$$\langle x, Ty\rangle = \langle T^*x, y\rangle.$$

An operator is self-adjoint when $T = T^*$, unitary when $T^*T = TT^* = I$, and normal when $T^*T = TT^*$. An *orthogonal projection* is an operator P satisfying $P = P^* = P^2$.

Definition 1.9 (Trace and Hilbert–Schmidt classes)**.** For $x, y \in H$, let

$$\theta_{x,y}(\xi) = \langle y, \xi\rangle x.$$

The algebra $\mathcal{K}(H)$ of *compact operators* is the norm closure of the finite-rank operators. The *trace class* $S_1(H)$ consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j,y_j}, \qquad \sum_{j=1}^{\infty} \|x_j\|\,\|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\mathrm{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j\rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^*A \in S_1(H)\}, \qquad \|A\|_2^2 = \mathrm{Tr}(A^*A).$$

Exercise 1.3 (Hilbert-space foundations)**.** Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \qquad (\operatorname{im} T)^\perp = \ker T^*.$$

For every closed subspace $M \subseteq H$, prove $H = M \oplus M^\perp$ and construct the orthogonal projection $P_M : H \to M$.

Exercise 1.4 (Trace ideals)**.** Prove that $S_1(H)$, Tr, and $S_2(H)$ are well-defined. Prove that $\langle A, B\rangle_{S_2} = \mathrm{Tr}(A^*B)$ makes $S_2(H)$ a Hilbert space and that the pairing $(A, T) \mapsto \mathrm{Tr}(AT)$ identifies $S_1(H)^*$ isometrically with $B(H)$.

Definition 1.10 (Fredholm determinant)**.** For $A \in S_1(H)$ with nonzero eigenvalues $\lambda_1(A), \lambda_2(A), \ldots$, repeated according to algebraic multiplicity, define

$$\det(I + zA) = \prod_{j=1}^{\infty} (1 + z\lambda_j(A)).$$

Exercise 1.5 (Fredholm determinant and trace identity)**.** Prove that the defining product converges absolutely and locally uniformly in z and agrees with the exterior-power determinant when A has finite rank. Prove

$$\det(I - zA) = \exp\left(-\sum_{m=1}^{\infty} \frac{z^m}{m}\,\mathrm{Tr}(A^m)\right)$$

for $|z| < \|A\|^{-1}$. Use the product to show that the determinant is entire and that its zeros are the reciprocals of the negatives of the nonzero eigenvalues of A. Construct the completed exterior powers of H and deduce

$$\det(I + zA) = \sum_{m=0}^{\infty} z^m\,\mathrm{Tr}(\Lambda^m A).$$

1.4 Measure, integration, and L^p spaces

Definition 1.11 (Measure spaces and L^p spaces)**.** A *σ-algebra* $\mathcal{F}$ on a set X contains $\varnothing$ and is closed under complements and countable unions. A *measure* is a countably additive map $\mu : \mathcal{F} \to [0, \infty]$ with $\mu(\varnothing) = 0$, and $(X, \mathcal{F}, \mu)$ is a *measure space*. A map is *measurable* when inverse images of measurable sets are measurable. For $1 \leq p < \infty$, define

$$L^p(X, \mu) = \left\{ f : \int_X |f|^p \, d\mu < \infty \right\} / \text{equality a.e.}$$

with its usual norm, and let $L^\infty(X, \mu)$ consist of essentially bounded measurable functions.

Definition 1.12 (Gamma and beta integrals)**.** For $s, t > 0$, define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} \, dx, \qquad B(s,t) = \int_0^1 x^{s-1}(1-x)^{t-1} \, dx.$$

Exercise 1.6 (Integration and L^p theorems)**.** Construct the Lebesgue integral from nonnegative simple functions. Prove the monotone convergence theorem, Fatou's lemma, the dominated convergence theorem, and Tonelli's and Fubini's theorems, stating the hypotheses that distinguish the last two. Prove Hölder's and Minkowski's inequalities and deduce that $L^p(X, \mu)$ is complete for $1 \leq p \leq \infty$. For $1 < p < \infty$ with $p^{-1} + q^{-1} = 1$, identify the continuous dual of L^p with L^q under the usual semifiniteness hypotheses.

Exercise 1.7 (Beta–gamma and Gaussian integral identities)**.** Use Tonelli's theorem and the change of variables $x = ru$, $y = r(1-u)$ to prove

$$B(s,t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}.$$

Use integration by parts to prove $\Gamma(s+1) = s\Gamma(s)$, and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \qquad \int_{\mathbb{R}^n} e^{-|x|^2/2} \, dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

1.5 C^*-algebras, functional calculus, and spectral decomposition

1.5.1 C*-algebras and functional calculus

Definition 1.13 (C^*-algebras, spectrum, and positivity)**.** A *C^*-algebra* is a complete normed complex algebra A with a conjugate-linear involution satisfying

$$(ab)^* = b^* a^*, \qquad \|ab\| \leq \|a\| \|b\|, \qquad \|a^* a\| = \|a\|^2.$$

For a unital C^*-algebra, define

$$\operatorname{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *positive*, written $a \geq 0$, when $a = b^* b$ for some $b \in A$. It is *self-adjoint* when $a = a^*$ and *normal* when $a^* a = a a^*$.

Definition 1.14 (Gelfand spectrum)**.** For a commutative C^*-algebra A, let $\widehat{A}$ be the space of nonzero homomorphisms $\chi : A \to \mathbb{C}$ with its weak-$*$ topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\widehat{A}), \qquad \Gamma(a)(\chi) = \chi(a).$$

Definition 1.15 (Positive functionals and states)**.** A linear functional $\varphi : A \to \mathbb{C}$ on a unital C^*-algebra is *positive* when $\varphi(a^*a) \geq 0$ for every $a \in A$. A *state* is a positive functional satisfying $\varphi(1) = 1$.

Exercise 1.8 (Spectrum, Gelfand duality, and functional calculus)**.** Prove that $\mathrm{spec}_A(a)$ is nonempty and compact and that

$$r(a) = \lim_{n\to\infty} \|a^n\|^{1/n}.$$

For normal a, prove $r(a) = \|a\|$. Prove that positive elements have unique positive square roots and that $a \geq 0$ if and only if $a = a^*$ and $\mathrm{spec}_A(a) \subseteq [0, \infty)$. Prove that the Gelfand transform is an isometric $*$-isomorphism. For every normal element a of a unital C^*-algebra, prove that there is a unique unital isometric $*$-homomorphism

$$C(\mathrm{spec}(a)) \longrightarrow C^*(1, a), \qquad f \longmapsto f(a),$$

sending the coordinate function to a, and prove the continuous spectral mapping theorem.

1.5.2 Spectral decomposition

Definition 1.16 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X is a map E from measurable subsets of X to orthogonal projections on H satisfying

$$E(\varnothing) = 0, \qquad E(X) = I, \qquad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral $\int_X f\, dE$ for bounded measurable f is defined by uniform approximation from simple functions. For a topological space X, the measure is *regular* when every scalar measure $B \mapsto \langle x, E(B)y\rangle$ is a regular Borel measure.

Exercise 1.9 (Representations of $C_0(X)$)**.** Let X be locally compact Hausdorff. A representation $\pi : C_0(X) \to B(H)$ is *nondegenerate* when $\overline{\pi(C_0(X))H} = H$. Prove that every nondegenerate $*$-representation has a unique regular projection-valued measure E_π satisfying

$$\pi(f) = \int_X f(x)\, dE_\pi(x), \qquad f \in C_0(X).$$

Exercise 1.10 (Bounded normal spectral theorem)**.** For a normal operator $T \in B(H)$, apply the preceding representation theorem to the continuous functional calculus $C(\mathrm{spec}(T)) \to B(H)$ and prove that there is a unique projection-valued measure E_T on $\mathrm{spec}(T)$ satisfying

$$T = \int_{\mathrm{spec}(T)} z\, dE_T(z), \qquad f(T) = \int_{\mathrm{spec}(T)} f(z)\, dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda\in\mathrm{spec}(T)} E_T(\{\lambda\})H, \qquad T|_{E_T(\{\lambda\})H} = \lambda I.$$

Exercise 1.11 (Compact spectral theorem)**.** Prove that every trace-class operator is compact. For a compact normal operator K, prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \operatorname{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm. For $K \geq 0$ in $S_1(H)$, prove

$$\operatorname{Tr} K = \sum_{\lambda \in \operatorname{spec}(K) \setminus \{0\}} \lambda \dim E_K(\{\lambda\}) H < \infty.$$

Definition 1.17 (Unbounded self-adjoint operators)**.** An unbounded operator is a linear map $A : \mathcal{D}(A) \to H$ on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax \rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For $y \in \mathcal{D}(A^*)$, the vector A^*y is determined by $\langle A^*y, x \rangle = \langle y, Ax \rangle$ for $x \in \mathcal{D}(A)$. The operator is *self-adjoint* when $A = A^*$, including equality of domains, and *closed* when its graph is closed in $H \oplus H$.

Exercise 1.12 (Unbounded spectral theorem and Stone's theorem)**.** For a self-adjoint operator A, prove that $A \pm iI$ are bijective from $\mathcal{D}(A)$ to H with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with $\ker(I - C_A) = 0$. Apply the bounded normal spectral theorem to C_A and the inverse Cayley map to prove that there is a unique projection-valued measure E_A on $\mathbb{R}$ satisfying

$$A = \int_{\mathbb{R}} \lambda \, dE_A(\lambda), \qquad \mathcal{D}(A) = \left\{ x : \int_{\mathbb{R}} \lambda^2 \, d\langle x, E_A(\lambda)x \rangle < \infty \right\}.$$

Use E_A to prove that e^{-itA} is a strongly continuous unitary group and that every strongly continuous one-parameter unitary group has a unique self-adjoint generator.

Exercise 1.13 (Spectral lines and time evolution)**.** Let A be self-adjoint, let $B \in B(H)$, and set $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$. For eigenvalues λ, μ of A, write $P_\lambda = E_A(\{\lambda\})$ and prove

$$P_\mu B(t) P_\lambda = e^{it(\mu - \lambda)/\hbar} P_\mu B P_\lambda.$$

Prove in the weak operator topology that

$$B(t) = \int_{\mathbb{R}} \int_{\mathbb{R}} e^{it(\mu - \lambda)/\hbar} \, dE_A(\mu) B \, dE_A(\lambda).$$

Deduce that the pure-point response measured through B can contain a spectral line at $(\mu - \lambda)/\hbar$ only when $P_\mu B P_\lambda \neq 0$, and identify vanishing spectral blocks as selection rules predicting absent lines.

Definition 1.18 (Airy function)**.** The Airy function is the oscillatory integral

$$\operatorname{Ai}(x) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{t^3}{3} + xt \right) dt.$$

Let $a_n > 0$ be determined by $\operatorname{Ai}(-a_n) = 0$, ordered increasingly.

Exercise 1.14 (Airy spectrum of the quantum bouncer)**.** On $L^2((0,\infty),dz)$ consider the self-adjoint Dirichlet realization

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dz^2} + mgz, \qquad \psi(0) = 0.$$

Prove that $\mathrm{Ai}''(x) = x\,\mathrm{Ai}(x)$ and that Ai decays as $x \to +\infty$. With

$$\ell_g = \left(\frac{\hbar^2}{2m^2 g}\right)^{1/3},$$

derive

$$E_n = mg\ell_g a_n, \qquad \psi_n(z) = C_n\,\mathrm{Ai}\left(\frac{z}{\ell_g} - a_n\right).$$

Use the spectral projections of $\widehat{H}$ to prove that a periodic perturbation $B\cos(\Omega t)$ can produce a transition from the nth to the kth level only through the block $E_{\widehat{H}}(\{E_k\})BE_{\widehat{H}}(\{E_n\})$, at the resonant frequency

$$\Omega_{kn} = \frac{mg\ell_g}{\hbar}|a_k - a_n|.$$

Deduce the transmission thresholds $z_n = \ell_g a_n$ for a horizontal absorber above a gravitationally bound beam and state how measured threshold heights and resonance frequencies determine g.

2 Group Symmetry and Abelian Harmonic Analysis

2.1 Groups, representations, and Haar integration

Definition 2.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Definition 2.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Definition 2.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V,\rho) \to (W,\sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 2.1 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 2.2 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 2.4 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h\rangle_G = \frac{1}{|G|}\sum_{g\in G}\overline{f(g)}h(g).$$

Exercise 2.3 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V \rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Definition 2.5 (Topological spaces)**.** A *topology* on X is a family of subsets containing $\varnothing, X$ and closed under arbitrary unions and finite intersections. A map is *continuous* when inverse images of open sets are open. A space is *Hausdorff* when distinct points have disjoint neighborhoods, *second countable* when its topology is generated by a countable family of open sets, *compact* when every open cover has a finite subcover, and *locally compact* when every point has a neighborhood with compact closure. The *Borel sets* form the σ-algebra generated by the open sets. A Borel measure is *regular* when it is inner regular on open sets and outer regular on Borel sets.

Definition 2.6 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

Definition 2.7 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$. The disjoint union $TM = \coprod_{p \in M} T_pM$ has a canonical smooth vector-bundle structure: if $x = (x^1, \ldots, x^n)$ is a coordinate chart on U, its bundle chart is

$$TM|_U \longrightarrow x(U) \times \mathbb{R}^n, \qquad v \in T_pM \longmapsto (x(p), v(x^1), \ldots, v(x^n)).$$

The cotangent bundle T^*M has the dual bundle charts.

Definition 2.8 (Locally compact groups and Haar measure)**.** A *topological group* is a group with a Hausdorff topology for which multiplication and inversion are continuous. It is *locally compact* when every point has a compact neighborhood. A *Radon measure* is a Borel measure finite on compact sets, outer regular on Borel sets, and inner regular on open sets. A *left Haar measure* on a locally compact group is a nonzero left-invariant Radon measure, so $\mu(gE) = \mu(E)$. Its *modular function* is determined by

$$\mu(Eg) = \Delta_G(g)^{-1}\mu(E).$$

Write $C_b(G)$, $C_0(G)$, and $C_c(G)$ for the continuous functions that are bounded, vanish at infinity, and have compact support, respectively.

Definition 2.9 (Unitary representations and matrix coefficients)**.** A *unitary representation* of a topological group G on a Hilbert space H_π is a homomorphism $\pi : G \to U(H_\pi)$ such that $g \mapsto \pi(g)x$ is continuous for every $x \in H_\pi$. Its *matrix coefficients* are

$$g \longmapsto \langle y, \pi(g)x \rangle, \qquad x, y \in H_\pi.$$

It is *irreducible* when it has no proper nonzero closed invariant subspace. The *unitary dual* $\widehat{G}$ is the set of unitary-equivalence classes of irreducible unitary representations.

Definition 2.10 (Lie groups)**.** A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 2.4 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 2.11 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \left.\frac{d}{dt}\right|_0 \rho(\exp tX).$$

Definition 2.12 (Convolution and the left regular representation)**.** Let G be a locally compact group with left Haar measure μ. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x)\, d\mu(y).$$

The *left regular representation* on $L^2(G)$ is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

Exercise 2.5 (Haar measure and the regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \le \|f\|_1 \|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 2.13 (Pontryagin dual)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\, d\mu(x).$$

The *double-dual map* sends $x \in G$ to the character $\chi \mapsto \chi(x)$ of $\widehat{G}$.

Exercise 2.6 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on $\mathbb{R}^n$ and Fourier series on $\mathbb{T}^n$. For a closed subgroup $H \le G$, define its annihilator $H^\perp \subseteq \widehat{G}$ and prove $\widehat{G/H} \cong H^\perp$.

2.2 Fourier analysis, lattices, and theta functions

2.2.1 Fourier series and Fourier transform

Definition 2.14 (Fourier series)**.** With normalized Haar measure on $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x)e^{-ikx}\, dx, \qquad k \in \mathbb{Z}.$$

The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}$.

Definition 2.15 (Schwartz space and convolution)**.** The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ consists of the smooth functions f satisfying

$$\sup_{x\in\mathbb{R}^n} |x^\alpha \partial^\beta f(x)| < \infty$$

for all multi-indices α, β. For integrable functions, define

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x-y)\,dy.$$

Definition 2.16 (Fourier transform)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, the semiclassical Fourier transform is

$$(\mathcal{F}_\hbar f)(p) = (2\pi\hbar)^{-n/2} \int_{\mathbb{R}^n} e^{-ip\cdot x/\hbar} f(x)\,dx.$$

Write $\mathcal{F} = \mathcal{F}_1$.

Exercise 2.7 (Fourier inversion and Plancherel)**.** Let G be a second-countable locally compact abelian group. Prove that every irreducible unitary representation is a character. Prove that Haar measure on $\widehat{G}$ can be normalized so that the group Fourier transform extends to a unitary map

$$L^2(G) \longrightarrow L^2(\widehat{G}),$$

and prove Fourier inversion when both f and $\widehat{f}$ are integrable. Prove that $(e^{ikx})_{k\in\mathbb{Z}}$ is a complete orthonormal family in $L^2(\mathbb{T})$ and derive

$$f = \sum_{k\in\mathbb{Z}} \widehat{f}(k)e^{ikx}, \qquad \|f\|_2^2 = \sum_{k\in\mathbb{Z}} |\widehat{f}(k)|^2.$$

Prove Fourier inversion on $\mathcal{S}(\mathbb{R}^n)$ and prove that $\mathcal{F}_\hbar$ extends uniquely to a unitary operator on $L^2(\mathbb{R}^n)$. Show that convolution becomes multiplication and differentiation becomes multiplication by frequency.

2.2.2 Lattices, Poisson summation, and diffraction

Definition 2.17 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi\cdot\gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Definition 2.18 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma\in\Gamma} f(x+\gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b\in B} c_b e^{-iq\cdot b}.$$

Exercise 2.8 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi\in\Gamma^*} \mathcal{F}f(\xi)e^{i\xi\cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 2.9 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq\cdot\gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d \sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

2.2.3 Lattice theta series and Poisson duality

Definition 2.19 (Integral lattices and theta series)**.** A *Euclidean lattice* $L \subset \mathbb{R}^d$ is a discrete subgroup for which $\mathbb{R}^d/L$ is compact. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y\rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if $L \subseteq L^*$, *unimodular* if $L = L^*$, and *even* if $\|x\|^2 \in 2\mathbb{Z}$ for every $x \in L$. For τ in the upper half-plane and $q = e^{2\pi i\tau}$, its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of q^n counts lattice vectors of squared length $2n$.

Exercise 2.10 (Poisson duality for lattice theta series)**.** Apply Exercise 2.8 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\mathrm{vol}(L)}\Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives $\Theta_L(\tau+1) = \Theta_L(\tau)$. Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

2.2.4 Modular forms, theta functions, and eta products

Definition 2.20 (The modular group and modular forms)**.** The modular group $SL_2(\mathbb{Z})$ acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. A *modular form of integral weight k* for $SL_2(\mathbb{Z})$ is a holomorphic function f on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion $f(\tau) = \sum_{n\geq 0} a_n q^n$ has no negative powers. It is a *cusp form* if $a_0 = 0$.

Definition 2.21 (Jacobi theta functions and the Dedekind eta function)**.** Set

$$\begin{aligned} \vartheta_2(\tau) &= \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad\qquad \eta(\tau) = q^{1/24}(q;q)_\infty. \\ \vartheta_4(\tau) &= \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2}, \end{aligned}$$

Exercise 2.11 (Theta transformations and the Jacobi triple product)**.** Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Prove the T-transformation laws directly from the series and derive

$$\eta(\tau+1) = e^{\pi i/12}\eta(\tau), \qquad \eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau),$$

using the holomorphic square root on the upper half-plane. Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2/2} = \prod_{m\geq 1}(1-q^m)(1+zq^{m-1/2})(1+z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

2.2.5 Eisenstein series and arithmetic applications

Definition 2.22 (Eisenstein series and the discriminant)**.** Let $\sigma_r(n) = \sum_{d|n} d^r$. The normalized Eisenstein series of weights four and six are

$$E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6(\tau) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n.$$

The modular discriminant is the weight-twelve cusp form

$$\Delta(\tau) = \eta(\tau)^{24} = q\prod_{n\geq 1}(1-q^n)^{24} = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728}.$$

Exercise 2.12 (Eisenstein series and the E_8 shell)**.** Construct Eisenstein series by summing $(m\tau+n)^{-k}$ over nonzero lattice points for even $k > 2$, prove their modular transformation law, and obtain the displayed Fourier expansions for E_4 and E_6. For a positive-definite even unimodular lattice, first use consistency of the S and T laws to prove that its dimension d is divisible by eight, and then show that its theta series is a modular form of weight $d/2$. Construct the E_8 lattice as

$$\{x \in \mathbb{Z}^8 : \textstyle\sum_i x_i \equiv 0 \pmod 2\} \cup \{x \in (\mathbb{Z}+\frac{1}{2})^8 : \textstyle\sum_i x_i \equiv 0 \pmod 2\}.$$

Enumerate its vectors of squared length two and obtain 240, matching the coefficient of q in E_4.

Exercise 2.13 (Theta heat trace and reciprocal-lattice diffraction)**.** Let $T_L = \mathbb{R}^d/L$ carry its flat metric, and normalize L^* by $\langle x, \xi\rangle \in \mathbb{Z}$. Prove

$$\mathrm{Tr}(e^{t\Delta_{T_L}}) = \sum_{\xi \in L^*} e^{-4\pi^2 t\|\xi\|^2} = \Theta_{L^*}(4\pi i t).$$

Apply Poisson duality to derive the short-time image sum and its leading term $\mathrm{vol}(T_L)(4\pi t)^{-d/2}$. For a periodic array with motif B, prove that the elastic intensity at reciprocal vector $2\pi\xi$ is proportional to

$$|S_B(2\pi\xi)|^2.$$

Prove that the heat-trace exponents and diffraction-peak radii determine the same reciprocal squared lengths $\|\xi\|^2$. Deduce the predicted peak locations, systematic absences, and thermal decay rates from the theta coefficients of L^*.

3 Compact Harmonic Analysis: Peter–Weyl, Characters, and Spin

Exercise 3.1 (Peter–Weyl theorem)**.** Let G be compact with normalized Haar measure. Prove that matrix coefficients of irreducible unitary representations are dense in $C(G)$ and $L^2(G)$ and that

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}}} H_\pi \otimes H_\pi^*.$$

Show that the left regular representation acts as $\pi \otimes I_{H_\pi^*}$ on the π-summand. Derive Schur orthogonality for irreducible matrix coefficients and the multiplicity $d_\pi = \dim H_\pi$. For $f \in L^2(G) \subseteq L^1(G)$, define $\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\,dg$ and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi)\,\mathrm{Tr}\big(\widehat{f}(\pi)\pi(\,\cdot\,)\big), \qquad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi)\|\widehat{f}(\pi)\|_{\mathrm{HS}}^2$$

, where the first sum converges in $L^2(G)$. For $f \in C(G)$, construct finite matrix-coefficient approximate identities and prove uniform convergence of the corresponding summability means to f.

Definition 3.1 (Rotations and spin)**.** Define

$$SO(3) = \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},$$
$$SU(2) = \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}.$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 3.2 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 3.3 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \ldots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \ldots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 3.4 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Exercise 3.5 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

and prove that the functions $Y_{\ell m} \propto D^{\ell}_{m0}$, $-\ell \leq m \leq \ell$, form a complete orthogonal family in V_ℓ. For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\phi^2.$$

Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$ and completeness of the spherical harmonics. Separate the Coulomb Hamiltonian in spherical coordinates, identify its angular factor, and recover $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi)$. Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

Exercise 3.6 (Central Fourier inversion and character identity)**.** Let G be compact with normalized Haar measure, and let f be a finite linear combination of irreducible characters. For an irreducible unitary representation π, set

$$\widehat{f}(\pi) = \int_G f(s)\pi(s^{-1})\,ds.$$

If f is central, use Schur's lemma to write $\widehat{f}(\pi) = c_\pi I_{d_\pi}$. Derive

$$f(e) = \sum_{\pi \in \widehat{G}} d_\pi \operatorname{Tr}(\widehat{f}(\pi)) = \sum_{\pi \in \widehat{G}} d_\pi^2 c_\pi$$

and prove that convolution by f acts by the scalar c_π on every copy of π in $L^2(G)$. Extend the identity to the natural Sobolev classes for which the character series converges absolutely.

Definition 3.2 (Clebsch–Gordan and Wigner coefficients)**.** For the normalized J_3-eigenstates in the irreducible angular-momentum representations, define

$$|jm\rangle = \sum_{m_1+m_2=m} C^{jm}_{j_1m_1,j_2m_2} |j_1m_1\rangle \otimes |j_2m_2\rangle,$$

with the highest-weight coefficient positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C^{j,-m_3}_{j_1m_1,j_2m_2}.$$

Define the Wigner $6j$ symbol by

$$|(j_1j_2)j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2j_3)j_{23}; jm\rangle.$$

Exercise 3.7 (Angular transition amplitudes and radiation patterns)**.** Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the $3j$ symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i}\, d\Omega = (-1)^{m_f} \sqrt{\frac{3(2\ell_i+1)(2\ell_f+1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules $\ell_f = \ell_i \pm 1$ and $m_f = m_i + q$, and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second $3j$ symbol after a common reduced matrix element is removed. Use the $6j$ identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines.

4 Semisimple Harmonic Analysis: Harish–Chandra Theory

Definition 4.1 (Cartan and Iwasawa decompositions)**.** Let G be a connected real semisimple Lie group with finite center and let $\mathfrak{g}$ be its Lie algebra. A *Cartan involution* gives a vector space decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

into its $+1$ and -1 eigenspaces; the subgroup K with Lie algebra $\mathfrak{k}$ is maximal compact, and G/K is a noncompact Riemannian symmetric space. Choose a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$, put $A = \exp \mathfrak{a}$, choose positive restricted roots, and let N be the subgroup generated by their root spaces. The *Iwasawa decomposition* is the diffeomorphism

$$K \times A \times N \longrightarrow G, \qquad (k, a, n) \longmapsto kan.$$

If $\mathfrak{a}^+$ is a positive Weyl chamber, the *Cartan decomposition* is $G = K\overline{A^+}K$. Write $M = Z_K(A)$ and $W = N_K(A)/M$, and let ρ be half the sum of the positive restricted roots, counted with multiplicity.

Definition 4.2 (Spherical principal series and spherical transform)**.** For $\lambda \in \mathfrak{a}^*$, the normalized *spherical principal series* is

$$\pi_\lambda = \mathrm{Ind}_{MAN}^G (1_M \otimes e^{i\lambda} \otimes 1_N),$$

where normalized induction incorporates the modular factor determined by ρ. It contains a unit K-fixed vector v_λ, and its *zonal spherical function* is

$$\varphi_\lambda(g) = \langle v_\lambda, \pi_\lambda(g) v_\lambda \rangle.$$

It is K-bi-invariant, satisfies $\varphi_\lambda(e) = 1$, and is a joint eigenfunction of the G-invariant differential operators on G/K.

For $f \in C_c^\infty(K\backslash G/K)$, its *spherical transform* is

$$\widetilde{f}(\lambda) = \int_G f(g) \varphi_{-\lambda}(g)\, dg.$$

Harish-Chandra's c-function is the coefficient governing the leading asymptotics of $\varphi_\lambda(\exp H)$ as $H \to \infty$ in a positive chamber.

Exercise 4.1 (Harish-Chandra transform on $\mathbb{H}^2$)**.** For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\mathrm{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta \varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For a smooth compactly supported radial function, define

$$\widetilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r) \sinh r \, dr.$$

Prove directly from the differential equation that distinct real values of λ^2 give orthogonal generalized eigenfunctions after truncation to $[0, R]$ with a common self-adjoint boundary condition. Compare the large-r equation with a constant-coefficient wave equation and show that its oscillation parameter is continuous. Contrast this with the discrete Dirichlet spectrum in the finite hyperbolic-drum exercise below.

Exercise 4.2 (A finite hyperbolic drum)**.** Discretize the Laplace–Beltrami operator by a real symmetric weighted graph Laplacian on a finite regular hyperbolic tiling with Dirichlet boundary. Prove that the graph Hilbert space is the orthogonal direct sum of the spectral subspaces of the finite operator, and explain why its spectrum approximates a bounded domain with a boundary rather than the continuous spectrum of all $\mathbb{H}^2$. Compare the predicted mode ordering and propagation along discrete geodesics with the circuit measurements in Lenggenhager et al., *Nature Communications* **13** (2022). State which observations support the effective hyperbolic metric and which do not by themselves establish the infinite-space Plancherel theorem.

Definition 4.3 (Harish-Chandra characters and discrete series)**.** A *distribution* on a Lie group G is a continuous linear functional on $C_c^\infty(G)$ with its usual test-function topology. Let π be an irreducible admissible unitary representation of a connected real reductive group. Although the infinite-dimensional operator $\pi(g)$ usually has no trace, for $f \in C_c^\infty(G)$ the integrated operator

$$\pi(f) = \int_G f(g)\pi(g) \, dg$$

is trace class. The *Harish-Chandra character* is the conjugation-invariant distribution

$$\Theta_\pi(f) = \mathrm{Tr}(\pi(f)).$$

An irreducible unitary representation belongs to the *discrete series* if it has a nonzero matrix coefficient square-integrable modulo the center.

Exercise 4.3 (From compact characters to the semisimple Plancherel formula)**.** For a compact group, prove

$$\Theta_\pi(f) = \int_G f(g)\,\mathrm{Tr}(\pi(g))\,dg$$

and recover the ordinary character used in Peter–Weyl theory. Show that a unitary operator on an infinite-dimensional Hilbert space is not trace class, which explains the distributional definition above. Prove directly that Θ_π is invariant under conjugation.

For $SL_2(\mathbb{R})$, identify the K-types of the spherical principal series and show that they contain a K-fixed vector. Contrast them with the holomorphic and antiholomorphic discrete series, whose nonzero one-sided K-types imply that they have no K-fixed vector. Deduce that spherical analysis on G/K sees the class-one principal series but not the discrete series. For $f \in C_c^\infty(K\backslash G/K)$, prove the spherical Plancherel and inversion formulas

$$\|f\|_2^2 = C_G \int_{\mathfrak{a}^*} |\widetilde{f}(\lambda)|^2 |c(\lambda)|^{-2}\,d\lambda, \qquad f(e) = C_G \int_{\mathfrak{a}^*} \widetilde{f}(\lambda)|c(\lambda)|^{-2}\,d\lambda,$$

with C_G determined by the Haar and Euclidean Fourier normalizations. For $SL_2(\mathbb{R})/SO(2)$, derive

$$c(\lambda) = C\frac{\Gamma(i\lambda)}{\Gamma(\frac{1}{2}+i\lambda)}, \qquad |c(\lambda)|^{-2} = C'\lambda\tanh(\pi\lambda),$$

with the constants fixed by the same normalization. Use spherical inversion to prove that the radial heat kernel and resolvent kernel on $\mathbb{H}^2$ are

$$p_t(r) = C_G \int_0^\infty e^{-t(\lambda^2+1/4)} P_{-1/2+i\lambda}(\cosh r)|c(\lambda)|^{-2}\,d\lambda,$$

$$G_\zeta(r) = C_G \int_0^\infty \frac{P_{-1/2+i\lambda}(\cosh r)}{\lambda^2+1/4-\zeta}|c(\lambda)|^{-2}\,d\lambda$$

away from the spectrum. Derive the return probability $p_t(0)$ and the high-frequency spectral density from $\lambda\tanh(\pi\lambda)$. For a hyperbolic wave network whose boundary is outside the observation region during the measurement interval, deduce the predicted radial attenuation and arrival profile of a localized pulse. State which discrepancies can be attributed to finite boundary, loss, and discretization before comparing the measured profile with the infinite-space kernel. Compare the two families tabulated in Harish-Chandra, *Proceedings of the National Academy of Sciences* **43** (1957) for spherical functions and Harish-Chandra, *Transactions of the American Mathematical Society* **119** (1965) for invariant eigendistributions.

5 Kirillov Orbit Method and the Heisenberg Group

5.1 Coadjoint orbits and the Kirillov form

Definition 5.1 (Coadjoint action and stabilizers)**.** For a Lie group G with Lie algebra $\mathfrak{g}$, define

$$\mathrm{Ad}_g = d(h \mapsto ghg^{-1})_e, \qquad (\mathrm{Ad}_g^*\,\ell)(X) = \ell(\mathrm{Ad}_{g^{-1}} X).$$

For $\ell \in \mathfrak{g}^*$, define

$$\mathcal{O}_\ell = \mathrm{Ad}^*(G)\ell, \qquad G_\ell = \{g \in G : \mathrm{Ad}_g^*\,\ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X \in \mathfrak{g} : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}.$$

For $X \in \mathfrak{g}$, set

$$(\operatorname{ad}_X^* \ell)(Y) = -\ell([X,Y]).$$

Definition 5.2 (Kirillov form)**.** On $T_\ell \mathcal{O}_\ell$, define

$$\omega_\ell(\operatorname{ad}_X^* \ell, \operatorname{ad}_Y^* \ell) = \ell([X,Y]).$$

For $\eta = \operatorname{Ad}_g^* \ell$, transport ω_ℓ by the coadjoint action to define ω_η.

Exercise 5.1 (Geometry of coadjoint orbits)**.** Prove

$$T_\ell \mathcal{O}_\ell = \{\operatorname{ad}_X^* \ell : X \in \mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and G-invariant. Prove the cyclic identity

$$\ell([X,[Y,Z]]) + \ell([Y,[Z,X]]) + \ell([Z,[X,Y]]) = 0$$

and identify it with the closedness identity for the invariant two-form ω on $\mathcal{O}_\ell$.

5.2 Polarizations, induction, and the Kirillov correspondence

Definition 5.3 (Polarizations and orbital representations)**.** Let G be connected, simply connected, and nilpotent. A *polarization* at $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{p} \subseteq \mathfrak{g}$ satisfying

$$\ell([\mathfrak{p},\mathfrak{p}]) = 0, \qquad \dim \mathfrak{p} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

For $P = \exp(\mathfrak{p})$, define

$$\chi_\ell(\exp X) = e^{i\ell(X)}.$$

The *orbital representation* $\pi_\ell = \operatorname{Ind}_P^G \chi_\ell$ acts on measurable functions $F : G \to \mathbb{C}$ satisfying

$$F(gp) = \chi_\ell(p)^{-1} F(g), \qquad \int_{G/P} |F(g)|^2 \, d(gP) < \infty,$$

by

$$(\pi_\ell(g_0)F)(g) = F(g_0^{-1} g),$$

where $d(gP)$ is a G-invariant quotient measure.

Exercise 5.2 (Kirillov correspondence)**.** Prove that every $\ell \in \mathfrak{g}^*$ admits a polarization, that $P = \exp(\mathfrak{p})$ is closed, and that χ_ℓ is a unitary character. Prove that π_ℓ is irreducible and independent of the polarization up to unitary equivalence. Prove

$$\pi_\ell \simeq \pi_{\ell'} \iff \mathcal{O}_\ell = \mathcal{O}_{\ell'}$$

and that

$$\mathfrak{g}^*/G \longrightarrow \widehat{G}, \qquad \mathcal{O}_\ell \longmapsto [\pi_\ell],$$

is a bijection.

Exercise 5.3 (Kirillov character formula)**.** Let f be a function on G for which $f \circ \exp \in \mathcal{S}(\mathfrak{g})$, choose dX so that $\exp_* dX$ is Haar measure, and define

$$\mathcal{F}_G f(\eta) = \int_{\mathfrak{g}} f(\exp X) e^{i\eta(X)}\, dX.$$

If $\dim \mathcal{O}_\ell = 2d$, define its orbital measure by

$$d\mu_{\mathcal{O}_\ell} = (2\pi)^{-d} \frac{\omega^d}{d!}.$$

For

$$\pi_\ell(f) = \int_G f(g) \pi_\ell(g)\, dg,$$

prove that $\pi_\ell(f)$ is trace class and that

$$\operatorname{Tr} \pi_\ell(f) = \int_{\mathcal{O}_\ell} \mathcal{F}_G f(\eta)\, d\mu_{\mathcal{O}_\ell}(\eta).$$

Compare the correspondence and character formula with Kirillov, *Russian Mathematical Surveys* **17** (1962).

5.3 Projective representations and multipliers

Definition 5.4 (Multipliers and projective representations)**.** A *normalized multiplier* on a locally compact group G is a continuous map $\sigma : G \times G \to \mathbb{T}$ satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \qquad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

A *σ-projective unitary representation* is a strongly continuous map $U : G \to U(H)$ satisfying

$$U_x U_y = \sigma(x, y) U_{xy}, \qquad U_e = I.$$

Multipliers σ and σ' are *cohomologous* when there is a continuous map $b : G \to \mathbb{T}$, $b(e) = 1$, such that

$$\sigma'(x, y) = b(x)b(y)\overline{b(xy)}\, \sigma(x, y).$$

5.4 Linear symplectic phase space

Definition 5.5 (Symplectic phase space)**.** A *symplectic form* on a finite-dimensional real linear space E is a nondegenerate alternating bilinear form $\omega : E \times E \to \mathbb{R}$. On $E = V \oplus V^*$, define

$$\omega\big((q, p), (q', p')\big) = p(q') - p'(q).$$

For $\lambda \in \mathbb{R}$, define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z,z')/2}.$$

The Weyl multiplier is σ_ω. For $a \in C^\infty(E)$, define $X_a : E \to E$ by

$$\omega(X_a(z), u) = -da_z(u) \qquad (z, u \in E),$$

and define the canonical Poisson bracket by

$$\{a, b\}(z) = -\omega(X_a(z), X_b(z)).$$

Exercise 5.4 (The Weyl multiplier)**.** Prove that $\sigma_{\lambda\omega}$ is a normalized multiplier for every $\lambda \in \mathbb{R}$.

5.5 Alternating commutator form

Definition 5.6 (Commutator bicharacter)**.** For a multiplier σ on an abelian group A, define

$$c_\sigma(x, y) = \sigma(x, y)\overline{\sigma(y, x)}.$$

Exercise 5.5 (Commutator bicharacter)**.** Prove that c_σ is an alternating bicharacter and depends only on the cohomology class of σ. For the Weyl multiplier on $V \oplus V^*$, prove

$$c_\sigma((q, p), (q', p')) = e^{i(p(q')-p'(q))}.$$

Exercise 5.6 (Linear Darboux theorem)**.** Prove that every finite-dimensional symplectic space is symplectically isomorphic to

$$(\mathbb{R}^{2n}, \omega_0), \qquad \omega_0((q, p), (q', p')) = p(q') - p'(q).$$

Prove

$$c_{\sigma_{\lambda\omega}}(z, z') = e^{i\lambda\omega(z,z')}.$$

5.6 Heisenberg group, Schrödinger representation, and CCR

Definition 5.7 (Heisenberg group)**.** For a symplectic space (E, ω), its *Heisenberg group* is

$$\mathbb{H}(E, \omega) = E \times \mathbb{R}$$

with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \tfrac{1}{2}\omega(z, z')).$$

Write $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$ for the canonical symplectic form on $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$. Its Lie algebra is $\mathfrak{h}(E, \omega) = E \oplus \mathbb{R}Z$ with

$$[(z, sZ), (z', s'Z)] = \omega(z, z')Z.$$

For $E = V \oplus V^*$ and $\lambda \neq 0$, the *Schrödinger representation* on $L^2(V)$ is

$$(\pi_\lambda(q, p, s)\xi)(x) = e^{i\lambda(s+p(x-q/2))}\xi(x - q).$$

For $\hbar > 0$, $\ell \in V^*$, and $v \in V$, define on $\mathcal{S}(V)$

$$Q_\ell\psi(x) = \ell(x)\psi(x), \qquad P_v\psi(x) = -i\hbar\, d\psi_x(v).$$

Exercise 5.7 (Central extensions and projective representations)**.** Prove that $\mathbb{H}(E, \omega)$ is a locally compact group with center $\{0\} \times \mathbb{R}$ and that

$$\pi_\lambda(0, s) = e^{i\lambda s}I.$$

Prove that projective representations of E with multiplier $e^{i\lambda\omega/2}$ correspond to representations of $\mathbb{H}(E, \omega)$ with central character $e^{i\lambda s}$.

Exercise 5.8 (Canonical commutation and Weyl relations)**.** Prove

$$[Q_\ell, P_v] = i\hbar\ell(v)I.$$

For

$$(T_q\psi)(x) = \psi(x - q), \qquad (M_p\psi)(x) = e^{ip(x)}\psi(x),$$

prove

$$M_pT_q = e^{ip(q)}T_qM_p.$$

For $W(q, p) = e^{-ip(q)/2}M_pT_q$, prove

$$W(z)W(z') = e^{i\omega(z,z')/2}W(z + z')$$

and identify W with the restriction of π_1 to $V \oplus V^* \subset \mathbb{H}(V \oplus V^*, \omega)$.

5.7 Heisenberg coadjoint orbits

Exercise 5.9 (Heisenberg orbits and the Schrödinger model)**.** Identify

$$\mathfrak{h}(E,\omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For $(\xi,\lambda) \in E^* \oplus \mathbb{R}Z^*$, prove

$$\mathrm{Ad}^*_{(z,s)}(\xi,\lambda) = (\xi - \lambda\omega(z,\cdot),\lambda).$$

Deduce that the orbits with $\lambda = 0$ are points and that, for $\lambda \neq 0$,

$$\mathcal{O}_\lambda = \{(\xi,\lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \qquad z \longmapsto (-\lambda\omega(z,\cdot),\lambda),$$

pulls the Kirillov form back to $\lambda\omega$.

For $E = V \oplus V^*$ and $\mathfrak{p} = V^* \oplus \mathbb{R}Z$, prove that $\mathfrak{p}$ is a polarization at $(0,\lambda)$. Identify $\mathbb{H}(E,\omega)/\exp(\mathfrak{p})$ with V and compute the induced action to obtain

$$(\pi_\lambda(q,p,s)\psi)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

For $\lambda = 0$, recover every character of E from a point orbit.

5.8 Stone–von Neumann theorem

Exercise 5.10 (Stone–von Neumann theorem)**.** Deduce from the Kirillov correspondence that π_λ is irreducible and that every strongly continuous irreducible unitary representation of $\mathbb{H}(V \oplus V^*,\omega)$ with central character $e^{i\lambda s}$, $\lambda \neq 0$, is unitarily equivalent to π_λ.

5.9 Fourier–Wigner transform

Definition 5.8 (Unitary systems and coefficient transforms)**.** Let (X,μ) be a measure space and let $U : X \to U(H)$ be a weakly measurable map. The *coefficient transform* of the unitary system U is

$$\mathcal{V}_U : S_1(H) \longrightarrow L^\infty(X), \qquad \mathcal{V}_U(A)(x) = \mathrm{Tr}(AU_x).$$

Definition 5.9 (Weyl system and Fourier–Wigner transform)**.** Let V be an n-dimensional real linear space with Lebesgue measure dq, and equip V^* with the dual measure dp for which $(2\pi)^{-n/2}\int_V e^{-ip(q)}f(q)\,dq$ is unitary on L^2. For $q \in V$ and $p \in V^*$, define

$$(T_q\xi)(x) = \xi(x-q), \qquad (M_p\xi)(x) = e^{ip(x)}\xi(x).$$

For $z = (q,p) \in V \oplus V^*$, define

$$W(z) = e^{-ip(q)/2}M_pT_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W} : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \qquad \mathcal{W}(A)(z) = (2\pi)^{-n/2}\,\mathrm{Tr}(AW(z)).$$

Exercise 5.11 (Fourier–Wigner Plancherel identity)**.** Prove

$$W(z)^* = W(-z).$$

For $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$, prove the Fourier–Wigner orthogonality identity

$$\int_{V \oplus V^*} \overline{\mathcal{W}(A)(z)} \mathcal{W}(B)(z)\, dz = \operatorname{Tr}(A^* B).$$

Deduce that $\mathcal{W}$ extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever $\mathcal{W}(A) \in L^1(V \oplus V^*)$, prove

$$A = (2\pi)^{-n/2} \int_{V \oplus V^*} \mathcal{W}(A)(z) W(z)^*\, dz$$

in the weak operator topology.

5.10 Heisenberg Plancherel theory

Exercise 5.12 (Heisenberg Plancherel and inversion)**.** For $V = \mathbb{R}^n$ and f satisfying $f \circ \exp \in \mathcal{S}(\mathfrak{h}_n)$, prove

$$\operatorname{Tr} \pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0,0,s) e^{i\lambda s}\, ds.$$

Derive this identity both from the Schrödinger kernel and from the Kirillov character formula on $\mathcal{O}_\lambda$. For $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$, use Haar measure $dq\, dp\, ds$ and the scalar Fourier transform $\widehat{h}(\lambda) = \int_{\mathbb{R}} h(s) e^{-i\lambda s}\, ds$, and define

$$\widehat{f}(\lambda) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g)^*\, dg.$$

Derive the Plancherel measure $(2\pi)^{-(n+1)} |\lambda|^n\, d\lambda$ and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R} \setminus \{0\}} \|\widehat{f}(\lambda)\|_{\mathrm{HS}}^2 |\lambda|^n\, d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

5.11 Hermite functions

Definition 5.10 (Hermite polynomials and functions)**.** Define the physicists' Hermite polynomials by

$$e^{2xt - t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2} H_n(x)}{(2^n n! \sqrt{\pi})^{1/2}}.$$

Exercise 5.13 (Hermite orthogonality and the Mehler kernel)**.** Prove that $(h_n)_{n\geq 0}$ is a complete orthonormal family in $L^2(\mathbb{R})$ and, for $|r|<1$, derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x)h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}}\exp\left(-\frac{(1+r^2)(x^2+y^2)-4rxy}{2(1-r^2)}\right).$$

Exercise 5.14 (Harmonic-oscillator spectroscopy, coherent states, and propagation)**.** For

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{m\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right), \qquad E_n = \hbar\omega\left(n+\frac{1}{2}\right)$$

give the complete spectral decomposition of $\widehat{H}$. Construct the creation and annihilation operators and compute

$$\langle \psi_m, Q\psi_n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n+1}\,\delta_{m,n+1} + \sqrt{n}\,\delta_{m,n-1}\right).$$

Deduce the spectral-line prediction $|E_{n\pm 1}-E_n|/\hbar = \omega$ and the relative line strengths for a linearly coupled oscillator. For $\alpha\in\mathbb{C}$, prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2}\sum_{n\geq 0}\frac{\alpha^n}{\sqrt{n!}}\psi_n,$$

show that its number statistics are Poisson with mean $|\alpha|^2$, and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual $t-i0$ prescription to obtain the oscillator propagator

$$K_t(x,y) = \sqrt{\frac{m\omega}{2\pi i\hbar\sin(\omega t)}}\exp\left[\frac{im\omega}{2\hbar\sin(\omega t)}\left((x^2+y^2)\cos(\omega t)-2xy\right)\right].$$

Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

Part II

Deformation

6 Geometry, Quantization, and Phase-Space Spectroscopy

6.1 Weyl quantization and the Moyal product

6.1.1 Weyl quantization and Wigner distributions

Definition 6.1 (Weyl quantization)**.** For $a\in\mathcal{S}(V\oplus V^*)$, define $\mathrm{Op}_\hbar^W(a):\mathcal{S}(V)\to\mathcal{S}(V)$ by

$$(\mathrm{Op}_\hbar^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n}\int_{V\times V^*} e^{ip(x-y)/\hbar}a\left(\frac{x+y}{2},p\right)\psi(y)\,dy\,dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on $\mathcal{S}(V)$.

Definition 6.2 (Wigner distributions)**.** For $\psi, \varphi \in \mathcal{S}(V)$, define

$$W_\hbar(\psi,\varphi)(q,p) = \frac{1}{(2\pi\hbar)^n}\int_V e^{-ip(y)/\hbar}\,\overline{\psi(q-y/2)}\,\varphi(q+y/2)\,dy.$$

Write $W_\hbar[\psi] = W_\hbar(\psi,\psi)$.

Exercise 6.1 (Weyl calculus and the Moyal product)**.** For $\ell \in V^*$ and $v \in V$, prove

$$\operatorname{Op}_\hbar^W((q,p)\mapsto \ell(q)) = Q_\ell, \qquad \operatorname{Op}_\hbar^W((q,p)\mapsto p(v)) = P_v.$$

Prove

$$\operatorname{Op}_\hbar^W(a)^* = \operatorname{Op}_\hbar^W(\overline{a}), \qquad \|\operatorname{Op}_\hbar^W(a)\|_{\mathrm{HS}} = (2\pi\hbar)^{-n/2}\|a\|_2,$$

and

$$\langle \psi, \operatorname{Op}_\hbar^W(a)\varphi\rangle = \int_{V\oplus V^*} a(q,p)W_\hbar(\psi,\varphi)(q,p)\,dq\,dp.$$

Prove that for $a, b \in \mathcal{S}(V\oplus V^*)$ there is a unique $a \star_\hbar b \in \mathcal{S}(V \oplus V^*)$ satisfying

$$\operatorname{Op}_\hbar^W(a\star_\hbar b) = \operatorname{Op}_\hbar^W(a)\operatorname{Op}_\hbar^W(b).$$

Derive the oscillatory-integral formula for $\star_\hbar$ and prove

$$a\star_\hbar b = ab + \frac{i\hbar}{2}\{a,b\} + O(\hbar^2), \qquad \frac{a\star_\hbar b - b\star_\hbar a}{i\hbar} = \{a,b\} + O(\hbar^2)$$

in the Schwartz-symbol topology.

Exercise 6.2 (Groenewold–van Hove obstruction)**.** Specialize to $V = \mathbb{R}$ and write $Q = Q_{\mathrm{id}_\mathbb{R}}$ and $P = P_1$. Prove that Weyl quantization carries Poisson brackets to $(i\hbar)^{-1}$ times commutators exactly for affine and quadratic polynomials. Quantize q^2p^2 using

$$q^2p^2 = \frac{1}{9}\{q^3,p^3\} = \frac{1}{3}\{q^2p, qp^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \qquad \mathcal{Q}(q) = Q, \qquad \mathcal{Q}(p) = P, \qquad \mathcal{Q}(q^m) = Q^m, \qquad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f,g\}) = (i\hbar)^{-1}[\mathcal{Q}(f),\mathcal{Q}(g)]$$

on a common invariant domain.

6.1.2 Laguerre phase-space functions

Definition 6.3 (Generalized Laguerre polynomials)**.** For $\alpha > -1$, define

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha}e^x}{n!}\frac{d^n}{dx^n}\left(e^{-x}x^{n+\alpha}\right), \qquad L_n = L_n^{(0)}.$$

Exercise 6.3 (Wigner–Laguerre identity)**.** Use the Laguerre Rodrigues formula to prove

$$\int_0^\infty L_m^{(\alpha)}(x)L_n^{(\alpha)}(x)x^\alpha e^{-x}\,dx = \frac{\Gamma(n+\alpha+1)}{n!}\delta_{mn}.$$

For

$$H_{\mathrm{cl}}(q,p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}, \qquad \psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}}q\right),$$

prove

$$W_\hbar[\psi_n](q,p) = \frac{(-1)^n}{\pi\hbar}e^{-2H_{\mathrm{cl}}(q,p)/(\hbar\omega)}L_n\left(\frac{4H_{\mathrm{cl}}(q,p)}{\hbar\omega}\right).$$

Verify the position and momentum marginals and identify the zeros and sign changes of the Wigner distribution from those of L_n.

6.2 Differential and symplectic geometry

6.2.1 Bundles, sections, and geometric structures

Definition 6.4 (Smooth bundles and sections)**.** A *smooth fiber bundle* with fiber F is a smooth map $\pi : E \to M$ locally isomorphic over M to $U \times F \to U$. A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal G-bundle* has a free right G-action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map $s : M \to E$ with $\pi s = \mathrm{id}_M$. For a representation $G \to \mathrm{GL}(V)$, the *associated vector bundle* is $P \times_G V = (P \times V)/((pg, v) \sim (p, gv))$.

Definition 6.5 (Frame bundles and G-structures)**.** The *frame bundle* of an n-manifold is the principal $\mathrm{GL}(n,\mathbb{R})$-bundle

$$\mathrm{Fr}(M)_p = \mathrm{Iso}(\mathbb{R}^n, T_pM).$$

For a closed subgroup $G \le \mathrm{GL}(n,\mathbb{R})$, a *$G$-structure* is a principal G-subbundle of $\mathrm{Fr}(M)$. It is a choice of the frames related by the transformations in G. Riemannian, Lorentzian, and almost symplectic structures correspond respectively to reductions to $O(n)$, $O(1, n-1)$, and $\mathrm{Sp}(2n,\mathbb{R})$; an oriented, time-oriented spacetime has structure group $SO^+(1,3)$.

Definition 6.6 (Tensor fields and metrics)**.** A (r,s)*-tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. Equivalently, an $O(p,q)$-structure is a smooth nondegenerate symmetric tensor g of signature (p,q). The cases $(n,0)$ and $(1,n-1)$ give Riemannian and Lorentzian metrics. The length of a Riemannian curve is $\int \sqrt{g(\dot\gamma,\dot\gamma)}\,dt$; the proper time of a timelike curve is $c^{-1}\int\sqrt{-g(\dot\gamma,\dot\gamma)}\,dt$. A metric and an orientation define a volume form vol_g.

6.2.2 Differential forms, cohomology, and Hodge theory

Definition 6.7 (Differential forms and exterior derivative)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$; write $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$. For a smooth map $F : M \to N$, define

$$(F^*\alpha)_p(v_1,\dots,v_k) = \alpha_{F(p)}(dF_p v_1,\dots,dF_p v_k).$$

The *exterior derivative* is the family $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and $F^*d = dF^*$.

Exercise 6.4 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

Definition 6.8 (Integration of forms)**.** An *orientation* of an n-manifold is a reduction of its frame bundle to $\mathrm{GL}^+(n,\mathbb{R})$, equivalently a continuous choice of component of nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 6.5 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt}\int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

Definition 6.9 (de Rham cohomology)**.** A differential form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d^2 = 0$, the *de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes $H^*_{\mathrm{dR}}(M)$ a graded-commutative algebra.

Exercise 6.6 (de Rham classes and periods)**.** On a star-shaped open subset of $\mathbb{R}^n$, construct a homotopy operator and prove the Poincaré lemma: every closed form of positive degree is exact. Use Stokes' theorem to prove that an exact k-form has zero integral over every closed oriented k-manifold mapped smoothly into M. Prove $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with the isomorphism given by integration.

On $\mathbb{R}^2 \setminus \{0\}$, show that

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed and has integral 2π around the unit circle, hence is not exact. Compute $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$. Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential $F = dA$ exactly when $[F] = 0$ in de Rham cohomology.

Definition 6.10 (Hodge star)**.** On an oriented pseudo-Riemannian n-manifold, the metric induces a nondegenerate pairing $\langle \cdot, \cdot \rangle_g$ on differential forms. The *Hodge star* is the bundle map

$$* : \Omega^p(M) \longrightarrow \Omega^{n-p}(M)$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle_g \ \mathrm{vol}_g$$

for p-forms α, β.

Exercise 6.7 (Metric volume and Hodge star)**.** Construct vol_g and the Hodge star intrinsically. For a metric with q negative directions, prove on p-forms that

$$*^2 = (-1)^{p(n-p)+q}\,\mathrm{id}\,.$$

6.2.3 Connections, curvature, and gauge fields

Definition 6.11 (Connections and parallel transport)**.** A *connection* on a vector bundle $E \to M$ is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fs) = X(f)s + f\nabla_X s$. Along a curve γ it induces a covariant derivative D/dt; its solutions $Ds/dt = 0$ define *parallel transport* between endpoint fibers. On TM, the torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$, and a *geodesic* satisfies $D\dot\gamma/dt = 0$.

Definition 6.12 (Curvature and holonomy)**.** The *curvature* of ∇ is the $\mathrm{End}(E)$-valued two-form

$$R^\nabla(X,Y)s = \nabla_X\nabla_Y s - \nabla_Y\nabla_X s - \nabla_{[X,Y]}s.$$

The *holonomy group* at p consists of the automorphisms of E_p obtained by parallel transport around loops based at p.

Exercise 6.8 (Parallel transport and curvature)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

Definition 6.13 (Principal connections and gauge fields)**.** Let $P \to M$ be a principal G-bundle. A *principal connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M. A reduction of the frame bundle describes spacetime geometry; a separate principal bundle describes an internal gauge symmetry.

Exercise 6.9 (Gauge covariance and Bianchi identity)**.** Prove $d_A F_A = 0$. For a gauge transformation Φ, prove $F_{\Phi^* A} = \Phi^* F_A$ and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy representation is trivial. For $G = U(1)$, compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

6.2.4 Symplectic manifolds and Hamiltonian reduction

Definition 6.14 (Symplectic manifolds and canonical transformations)**.** A *symplectic form* is a closed nondegenerate two-form ω. For $f \in C^\infty(M)$, define

$$\iota_{X_f}\omega = -df, \qquad \{f,g\} = -\omega(X_f, X_g).$$

A diffeomorphism $F : (M,\omega_M) \to (N,\omega_N)$ is a *canonical transformation* when $F^*\omega_N = \omega_M$. For a manifold Q, the tautological one-form on T^*Q is

$$\theta_\alpha(v) = \alpha(d\pi(v)),$$

and the canonical symplectic form is $\omega = d\theta$.

Definition 6.15 (Classical states and observables)**.** A *pure classical state* is a point of M, a *classical observable* is a function $f \in C^\infty(M,\mathbb{R})$, and a *statistical state* is a Borel probability measure μ with expectation

$$\mathbb{E}_\mu[f] = \int_M f\, d\mu.$$

For Hamiltonian flow Φ_t, define

$$\mu_t = (\Phi_t)_* \mu_0, \qquad f_t = f \circ \Phi_t.$$

Exercise 6.10 (Darboux theorem and Hamiltonian mechanics)**.** Prove that every point of a $2n$-dimensional symplectic manifold has a neighborhood symplectomorphic to an open subset of $(\mathbb{R}^{2n}, \omega_0)$. Identify the canonical symplectic form on T^*Q from the tautological one-form. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. Prove

$$\frac{d}{dt} f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

Definition 6.16 (Hamiltonian actions and reduction)**.** A symplectic action of a Lie group G on (M, ω) is *Hamiltonian* when it has an equivariant map $J : M \to \mathfrak{g}^*$ satisfying

$$d\langle J, \xi \rangle = -\iota_{\xi_M} \omega \qquad (\xi \in \mathfrak{g}).$$

Let $\eta \in \mathfrak{g}^*$ be a regular value and let $G_\eta = \{g : \mathrm{Ad}_g^* \eta = \eta\}$. If G_η acts freely and properly on $J^{-1}(\eta)$, define

$$M//_\eta G = J^{-1}(\eta)/G_\eta, \qquad q^* \omega_\eta = i^* \omega,$$

where $i : J^{-1}(\eta) \hookrightarrow M$ and $q : J^{-1}(\eta) \to M//_\eta G$.

Exercise 6.11 (Noether and Marsden–Weinstein theorems)**.** Prove that the components of J Poisson-commute with every invariant Hamiltonian. Prove that ω_η exists uniquely and is symplectic and that invariant Hamiltonian flows descend to $M//_\eta G$. Compute the momentum maps for translations and rotations on $T^*\mathbb{R}^3$.

Exercise 6.12 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the volume form $\omega^n/n!$, and every density depending only on the Hamiltonian.

Exercise 6.13 (Pendulum periods and the complete elliptic integral)**.** On T^*S^1 consider

$$H(\theta, p) = \frac{p^2}{2m\ell^2} + mg\ell(1 - \cos\theta).$$

For an oscillation of amplitude $0 < \theta_0 < \pi$, set $k = \sin(\theta_0/2)$ and define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Use energy conservation to prove the exact period identity

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}}\, K(k).$$

Expand the integral to obtain

$$T(\theta_0) = 2\pi \sqrt{\frac{\ell}{g}} \left(1 + \frac{\theta_0^2}{16} + O(\theta_0^4)\right).$$

Deduce the experimentally testable increase of period with amplitude and the logarithmic divergence as $\theta_0 \uparrow \pi$.

6.3 Classical dynamics and geometric quantization

6.3.1 Classical and quantum evolution

Definition 6.17 (Classical and quantum evolution)**.** Let $H \in C^\infty(V \oplus V^*, \mathbb{R})$ have bounded derivatives of order at least two, let Φ_t be its Hamiltonian flow, and let $\mathrm{Op}_\hbar^W(H)$ denote its self-adjoint realization. Define

$$U_\hbar(t) = e^{-it\,\mathrm{Op}_\hbar^W(H)/\hbar}.$$

For $a \in \mathcal{S}(V \oplus V^*)$, define

$$a_t = a \circ \Phi_t, \qquad A_\hbar(t) = U_\hbar(t)^* \,\mathrm{Op}_\hbar^W(a) U_\hbar(t).$$

Exercise 6.14 (Heisenberg equation and Egorov's theorem)**.** Prove

$$\frac{d}{dt} a_t = \{a_t, H\}, \qquad \frac{d}{dt} A_\hbar(t) = \frac{i}{\hbar}[\mathrm{Op}_\hbar^W(H), A_\hbar(t)].$$

For every $T > 0$, prove Egorov's theorem

$$\sup_{|t| \le T} \left\| A_\hbar(t) - \mathrm{Op}_\hbar^W(a \circ \Phi_t) \right\| = O_T(\hbar).$$

Prove exact equality for quadratic H. Derive Ehrenfest's equations for the expectations of Q_ℓ and P_v for every $\ell \in V^*$ and $v \in V$.

6.3.2 Prequantization and polarization

Definition 6.18 (Prequantum line bundles)**.** A *prequantum line bundle* over (M, ω) is a Hermitian line bundle $L \to M$ with a unitary connection ∇ satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega.$$

For $f \in C^\infty(M)$, its *Kostant–Souriau operator* is

$$\mathcal{Q}_{\mathrm{pre}}(f) = -i\hbar \nabla_{X_f} + f.$$

Definition 6.19 (Polarizations and metaplectic correction)**.** A *complex polarization* is an involutive Lagrangian subbundle $P \subset T_{\mathbb{C}}M$. Its canonical line is $K_P = \Lambda^{\mathrm{top}} P^*$, and a *metaplectic correction* is a line bundle δ_P with $\delta_P^{\otimes 2} \cong K_P$. The corrected quantum space is the Hilbert completion of sections of $L \otimes \delta_P$ covariantly constant along P.

Exercise 6.15 (Prequantization and polarization)**.** Prove that a prequantum line bundle exists if and only if

$$\left[\frac{\omega}{2\pi\hbar}\right] \in H^2(M; \mathbb{Z}).$$

Prove

$$[\mathcal{Q}_{\mathrm{pre}}(f), \mathcal{Q}_{\mathrm{pre}}(g)] = i\hbar \mathcal{Q}_{\mathrm{pre}}(\{f, g\}).$$

For T^*V with the vertical polarization, recover $L^2(V)$ and the operators Q_ℓ and P_v. For a compact Kähler manifold with a positive prequantum line bundle, identify the corrected quantum space with $H^0(M, L \otimes K_M^{1/2})$.

6.3.3 Equivariant quantization and reduction

Definition 6.20 (Equivariant quantization)**.** Let a Lie group G act on a prequantized Hamiltonian manifold $(M, \omega, J, L, \nabla)$ and preserve a polarization P and metaplectic correction δ_P. An *equivariant quantization* is a lift of the G-action to $L \otimes \delta_P$ preserving its Hermitian structure, connection, and polarization. Its quantum representation is the induced unitary representation on the polarized quantum space.

Exercise 6.16 (Quantization commutes with reduction)**.** For a compact connected group acting holomorphically and Hamiltonianly on a compact prequantized Kähler manifold, assume that 0 is a regular value of J, that G acts freely on $J^{-1}(0)$, and that the prequantum and half-form data descend to the quotient. Prove that the derived quantum representation is generated by the quantized components of J and prove

$$\mathcal{Q}(M//_0G) \cong \mathcal{Q}(M)^G$$

with the metaplectic correction. Derive the multiplicity formula obtained by reduction at integral coadjoint orbits.

6.4 Deformation quantization

Definition 6.21 (Poisson manifolds)**.** A *Poisson manifold* is a smooth manifold M with a bracket on $C^\infty(M)$ that is a Lie bracket and a derivation in each argument. Equivalently, there is a bivector $\Pi \in \Gamma(\Lambda^2 TM)$ satisfying

$$\{f, g\} = \Pi(df, dg), \qquad [\Pi, \Pi]_{\mathrm{Sch}} = 0.$$

Definition 6.22 (Star products)**.** A *differential star product* on a Poisson manifold is an associative $\mathbb{C}[[\hbar]]$-bilinear product on $C^\infty(M)[[\hbar]]$ of the form

$$f \star g = fg + \sum_{r \geq 1} \hbar^r C_r(f, g),$$

where the C_r are bidifferential operators, 1 is a unit, and

$$C_1(f, g) - C_1(g, f) = i\{f, g\}.$$

Two star products are *equivalent* when they are intertwined by an operator $T = \mathrm{id} + \sum_{r \geq 1} \hbar^r T_r$ with differential T_r.

Exercise 6.17 (Fedosov and Kontsevich quantization)**.** Prove that the Moyal product is a differential star product on $\mathbb{R}^{2n}$. Prove that every symplectic manifold admits a Fedosov star product and that its equivalence class is determined by its formal characteristic class. Prove the Kontsevich formality theorem and deduce that every finite-dimensional Poisson manifold admits a deformation quantization.

Exercise 6.18 (Moyal plane-wave identity)**.** On $\mathbb{R}^{2n}$ let Π be the canonical Poisson tensor and write $\Pi(k, \ell) = k_i \Pi^{ij} \ell_j$ for covectors k, ℓ. Derive from the Moyal product

$$e^{ik\cdot z} \star_\hbar e^{i\ell\cdot z} = e^{-i\hbar\Pi(k,\ell)/2} e^{i(k+\ell)\cdot z}.$$

Prove associativity directly by checking the multiplier identity for the phase. Differentiate in k and ℓ to recover the canonical commutation relations and the full bidifferential expansion of the Moyal product on polynomials.

Exercise 6.19 (Laguerre signatures in phase-space tomography)**.** For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}} Q \cos\theta + \frac{P}{\sqrt{m\hbar\omega}} \sin\theta,$$

prove that its probability density in a state ψ is the Radon transform of $W_\hbar[\psi]$ along the corresponding phase-space lines. Derive the Fourier-slice inversion formula reconstructing the Wigner distribution from the family of quadrature densities. For the oscillator state ψ_n, use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi\hbar} e^{-2H_{\mathrm{cl}}/(\hbar\omega)} L_n\left(\frac{4H_{\mathrm{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled $e^{-x^2} H_n(x)^2$. Deduce the predicted number of quadrature nodes, the radial zero circles determined by L_n, and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction.

7 Integrable Systems and the Yang–Baxter Equation

Definition 7.1 (Liouville integrability and Lax pairs)**.** A Hamiltonian system on a $2n$-dimensional symplectic manifold is *Liouville integrable* when it has functions $F_1 = H, F_2, \ldots, F_n$ that are independent on a dense open set and satisfy $\{F_i, F_j\} = 0$. A *Lax representation* is an equation

$$\dot{L} = [M, L]$$

for matrix- or operator-valued functions L and M on phase space.

Definition 7.2 (Spectral Yang–Baxter equation)**.** Let V be finite-dimensional. A meromorphic family $R(u) \in \mathrm{End}(V \otimes V)$ satisfies the *spectral Yang–Baxter equation* when

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v)$$

where both sides are defined. It is *regular* when $R(0)$ is a nonzero scalar multiple of the flip and *unitary* when $R_{12}(u)R_{21}(-u)$ is scalar.

Exercise 7.1 (Lax invariants and factorized scattering)**.** For a Lax equation, prove

$$\frac{d}{dt} \operatorname{Tr}(L^r) = 0 \qquad (r \geq 1)$$

and that the characteristic polynomial of L is constant. Interpret the spectral Yang–Baxter equation as equality of the two factorizations of a three-body scattering operator. Show that regularity and unitarity give the identity and inverse moves needed for consistent factorized scattering.

Exercise 7.2 (A direct optical Yang–Baxter test)**.** Consider the unitary polarization transformations

$$A(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \qquad B(\theta) = \begin{pmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{pmatrix}.$$

Multiply the matrices and prove that, away from singular boundary cases,

$$A(\theta_1)B(\theta_2)A(\theta_3) = B(\theta_3)A(\theta_2)B(\theta_1)$$

exactly when

$$\tan\theta_2 = \frac{\sin(\theta_1+\theta_3)}{\cos(\theta_1-\theta_3)}.$$

For $\theta_1 = 56^\circ$ and $\theta_3 = 23^\circ$, obtain $\theta_2 \approx 49.49^\circ$. For an arbitrary input qubit $|\psi\rangle$, show that the output-state overlap has absolute value one when the relation holds and is generally smaller away from it. Compare these predictions with the wave-plate experiment of Zheng et al., *Journal of the Optical Society of America B* **30** (2013). State clearly that this experiment tests a two-dimensional Temperley–Lieb reduction of the Yang–Baxter equation, not the full representation theory of a quantum group or DAHA.

8 Rational Hypergeometric Functions

Definition 8.1 (Ordinary hypergeometric series)**.** For $m \in \mathbb{Z}_{\geq 0}$, define

$$(a)_0 = 1, \qquad (a)_m = a(a+1)\cdots(a+m-1) \quad (m \geq 1).$$

Where the quotient is defined, $(a)_m = \Gamma(a+m)/\Gamma(a)$; the product gives its continuation at nonpositive integral a. If no denominator parameter b_j is a nonpositive integer, the *generalized hypergeometric series* is

$${}_rF_s\left(\begin{matrix} a_1,\dots,a_r \\ b_1,\dots,b_s \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a_1)_m\cdots(a_r)_m}{(b_1)_m\cdots(b_s)_m}\frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer. Otherwise it converges for every z when $r \leq s$, for $|z| < 1$ when $r = s+1$, and only at $z = 0$ when $r > s+1$. The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z)\frac{d^2}{dz^2} + (c-(a+b+1)z)\frac{d}{dz} - ab.$$

Exercise 8.1 (Gauss hypergeometric theorem)**.** Prove

$${}_1F_0\left(\begin{matrix} a \\ - \end{matrix}; z\right) = (1-z)^{-a}$$

and prove that ${}_2F_1(a,b;c;z)$ is the solution of $\mathcal{L}_{a,b,c}f = 0$ analytic at $z = 0$ with $f(0) = 1$. Derive Euler's integral and transformation

$${}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)}\int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a}\,du,$$

$${}_2F_1(a,b;c;z) = (1-z)^{c-a-b}{}_2F_1(c-a,c-b;c;z),$$

and Gauss's summation

$${}_2F_1(a,b;c;1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}.$$

State the parameter conditions for each identity and extend them by analytic continuation.

Definition 8.2 (Bessel functions)**.** For $\alpha > -1$, the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!\,\Gamma(m+\alpha+1)} \left(\frac{z}{2}\right)^{2m+\alpha},$$

with a branch of z^α fixed when $\alpha \notin \mathbb{Z}$.

Exercise 8.2 (Bessel equation and radial Fourier transform)**.** Prove

$$z^2 J_\alpha''(z) + zJ_\alpha'(z) + (z^2 - \alpha^2)J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z} J_\alpha(z), \qquad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on $\mathbb{R}^n$ as a Hankel transform with kernel $J_{n/2-1}$. Derive the corresponding inversion identity.

Definition 8.3 (Jacobi and Wilson polynomials)**.** For $\alpha, \beta > -1$, the *Jacobi polynomial* is

$$P_n^{(\alpha,\beta)}(x) = \frac{(\alpha+1)_n}{n!} {}_2F_1\left(\begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2}\right).$$

For positive a, b, c, d, the *Wilson polynomial* is

$$\begin{aligned} W_n(x^2; a, b, c, d) &= (a+b)_n(a+c)_n(a+d)_n \\ &\quad \times {}_4F_3\left(\begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1\right). \end{aligned}$$

Exercise 8.3 (Rational rank-one spectral theory)**.** Prove

$$\left((1-x^2)\frac{d^2}{dx^2} + (\beta - \alpha - (\alpha+\beta+2)x)\frac{d}{dx}\right) P_n^{(\alpha,\beta)} = -n(n+\alpha+\beta+1)P_n^{(\alpha,\beta)}.$$

Derive Jacobi orthogonality on $[-1, 1]$ with weight $(1-x)^\alpha(1+x)^\beta$. Prove the Wilson difference equation and orthogonality on $[0, \infty)$ with weight

$$w_W(x) = \left|\frac{\Gamma(a+ix)\Gamma(b+ix)\Gamma(c+ix)\Gamma(d+ix)}{\Gamma(2ix)}\right|^2.$$

Derive the three-term recurrences of both families.

Exercise 8.4 (Bessel resonances and circular diffraction)**.** Let $j_{\nu,k}$ be the kth positive zero of J_ν. Separate the wave equation on a circular membrane of radius R with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R} j_{|m|,k},$$

with radial factor $J_{|m|}(j_{|m|,k}r/R)$. Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of c/R.

For a uniformly illuminated circular aperture of radius a, average the Fraunhofer kernel over the aperture and derive

$$\frac{I(\theta)}{I(0)} = \left[\frac{2J_1(ka\sin\theta)}{ka\sin\theta}\right]^2.$$

Prove that the dark-ring angles satisfy $ka\sin\theta_k = j_{1,k}$ and obtain the small-angle prediction $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$. Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

9 Quantum Groups and Basic Hypergeometric Spectra

9.1 Basic hypergeometric functions

Definition 9.1 (Basic hypergeometric series)**.** For $m \in \mathbb{N}$, define

$$(a;q)_m = \prod_{j=0}^{m-1}(1-aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1-aq^j),$$

and

$$(a_1,\dots,a_r;q)_m = \prod_{i=1}^{r}(a_i;q)_m, \qquad (a_1,\dots,a_r;q)_\infty = \prod_{i=1}^{r}(a_i;q)_\infty.$$

The *basic hypergeometric series* is

$${}_r\phi_s\left(\begin{matrix}a_1,\dots,a_r\\ b_1,\dots,b_s\end{matrix};q,z\right) = \sum_{m=0}^{\infty}\frac{(a_1,\dots,a_r;q)_m}{(q,b_1,\dots,b_s;q)_m}\left((-1)^m q^{m(m-1)/2}\right)^{1+s-r} z^m.$$

It is *terminating* when one numerator parameter is q^{-n}.

Exercise 9.1 (Basic hypergeometric summations)**.** Prove the q-binomial theorem

$${}_1\phi_0\left(\begin{matrix}a\\ -\end{matrix};q,z\right) = \frac{(az;q)_\infty}{(z;q)_\infty}, \qquad |z|<1,$$

the q-Gauss summation

$${}_2\phi_1\left(\begin{matrix}a,b\\ c\end{matrix};q,\frac{c}{ab}\right) = \frac{(c/a,c/b;q)_\infty}{(c,c/(ab);q)_\infty}, \qquad \left|\frac{c}{ab}\right|<1,$$

and the terminating q-Pfaff–Saalschütz summation

$${}_3\phi_2\left(\begin{matrix}a,b,q^{-n}\\ c,abq^{1-n}/c\end{matrix};q,q\right) = \frac{(c/a,c/b;q)_n}{(c,c/(ab);q)_n}.$$

9.2 Poisson–Lie groups and quantum groups

9.2.1 Poisson–Lie groups and Lie bialgebras

Definition 9.2 (Poisson–Lie groups and Lie bialgebras)**.** A *Poisson–Lie group* is a Lie group G with a Poisson structure for which multiplication $G \times G \to G$ is Poisson. A *Lie bialgebra* is a Lie algebra $\mathfrak{g}$ with a linear map

$$\delta : \mathfrak{g} \to \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on $\mathfrak{g}^*$. It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some $r \in \mathfrak{g} \otimes \mathfrak{g}$.

Exercise 9.2 (Infinitesimal Poisson–Lie duality)**.** Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. Prove the converse for connected simply connected groups. For a coboundary Lie bialgebra, define

$$\mathrm{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Prove that the co-Jacobi identity is equivalent to $\mathrm{CYB}(r)$ being invariant under the diagonal adjoint action. Under the additional triangular hypothesis $\mathrm{CYB}(r) = 0$, recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as $\mathrm{CYB}(r) = \Omega$ for an invariant tensor Ω and distinguish the case $\Omega \neq 0$.

9.2.2 Drinfeld–Jimbo quantization

Definition 9.3 (Bialgebras and Hopf algebras)**.** A *bialgebra* is a unital associative algebra H with unital algebra homomorphisms

$$\Delta : H \to H \otimes H, \qquad \varepsilon : H \to \mathbb{C}$$

satisfying coassociativity and the counit identities. A *Hopf algebra* is a bialgebra with a linear antipode $S : H \to H$ satisfying

$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta.$$

Definition 9.4 (Quantized enveloping algebras)**.** A *quantized enveloping algebra* of $(\mathfrak{g}, \delta)$ is a topologically free Hopf algebra $U_\hbar(\mathfrak{g})$ over $\mathbb{C}[[\hbar]]$ satisfying

$$U_\hbar(\mathfrak{g})/\hbar U_\hbar(\mathfrak{g}) \cong U(\mathfrak{g}), \qquad \delta(x) = \frac{\Delta_\hbar(x) - \Delta_\hbar^{\mathrm{op}}(x)}{\hbar} \bmod \hbar.$$

For generic q, $U_q(\mathfrak{sl}_2)$ is generated by E, F, K, K^{-1} with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F, \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Exercise 9.3 (Drinfeld–Jimbo theorem)**.** Construct $U_q(\mathfrak{g})$ from the Cartan matrix of a complex semisimple Lie algebra and prove that it is a Hopf deformation of $U(\mathfrak{g})$. For $\mathfrak{sl}_2$, verify the displayed relations and coproduct and compute the semiclassical Lie bialgebra.

Exercise 9.4 (Quantum-group binomial identity)**.** For $YX = q^2 XY$, define

$$\begin{bmatrix} n \\ r \end{bmatrix}_{q^2} = \frac{(q^2; q^2)_n}{(q^2; q^2)_r (q^2; q^2)_{n-r}}.$$

Prove the noncommutative binomial identity

$$(X + Y)^n = \sum_{r=0}^{n} \begin{bmatrix} n \\ r \end{bmatrix}_{q^2} X^r Y^{n-r}.$$

Apply it to $X = E \otimes 1$ and $Y = K \otimes E$ in $U_q(\mathfrak{sl}_2)$ to obtain an explicit formula for $\Delta(E^n)$. Show that the limit $q \to 1$ recovers the ordinary binomial theorem and the undeformed coproduct.

9.3 Compact quantum groups

Definition 9.5 (Compact quantum groups and Haar states)**.** A *compact quantum group* $\mathbb{G}$ is a unital C^*-algebra $C(\mathbb{G})$ with a unital $*$-homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where $\otimes_{\min}$ is the closure of the algebraic tensor product in $B(H \otimes K)$ for faithful representations on H and K. The coproduct satisfies

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$$

and

$$\overline{\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))} = C(\mathbb{G}) \otimes_{\min} C(\mathbb{G}) = \overline{\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)}.$$

A *Haar state* is a state h satisfying

$$(h \otimes \mathrm{id})\Delta(a) = h(a)1 = (\mathrm{id} \otimes h)\Delta(a).$$

Definition 9.6 (Corepresentations and representative functions)**.** A finite-dimensional *unitary corepresentation* on H_U is a unitary $U \in B(H_U) \otimes C(\mathbb{G})$ satisfying

$$(\mathrm{id} \otimes \Delta)(U) = U_{12}U_{13}.$$

For $\omega \in B(H_U)^*$, the corresponding *coefficient* is

$$u_\omega = (\omega \otimes \mathrm{id})(U) \in C(\mathbb{G}).$$

The representative-function algebra $\mathrm{Pol}(\mathbb{G})$ is the unital $*$-subalgebra generated by these coefficients for all finite-dimensional unitary corepresentations.

Definition 9.7 (Drinfeld–Jimbo and Woronowicz duality)**.** Let $0 < q < 1$, let G be a compact connected simply connected semisimple Lie group, and let $\mathfrak{g}_\mathbb{C}$ be its complexified Lie algebra. The Hopf $*$-algebra $\mathrm{Pol}(G_q)$ is the representative-function algebra of finite-dimensional type-1 unitary $U_q(\mathfrak{g}_\mathbb{C})$-modules for the compact real-form involution, with pairing

$$\langle x, c_{\varphi,v}\rangle = \varphi(xv).$$

Its universal and reduced C^*-completions are denoted $C_u(G_q)$ and $C_r(G_q)$.

Exercise 9.5 (Woronowicz Peter–Weyl theorem)**.** Prove existence and uniqueness of the Haar state. For $\alpha \in \widehat{\mathbb{G}}$, let C_α be its coefficient space and prove the algebraic Peter–Weyl decomposition

$$\mathrm{Pol}(\mathbb{G}) = \bigoplus_{\alpha \in \widehat{\mathbb{G}}} C_\alpha.$$

Prove that $\mathrm{Pol}(\mathbb{G})$ is a dense Hopf $*$-subalgebra of $C(\mathbb{G})$. For each irreducible U^α, construct the normalized positive invertible operator $Q_\alpha \in B(H_{U^\alpha})$ satisfying

$$(\mathrm{id} \otimes S^2)(U^\alpha) = (Q_\alpha \otimes 1)U^\alpha(Q_\alpha^{-1} \otimes 1), \qquad \mathrm{Tr}(Q_\alpha) = \mathrm{Tr}(Q_\alpha^{-1}),$$

and derive the orthogonality relations in the Haar GNS space and the resulting Hilbert direct-sum decomposition. Prove that the displayed pairing between $U_q(\mathfrak{g}_\mathbb{C})$ and $\mathrm{Pol}(G_q)$ is a nondegenerate Hopf pairing.

Exercise 9.6 ($SU_q(2)$)**.** Let $C(SU_q(2))$ be generated by α, γ so that

$$u = \begin{pmatrix} \alpha & -q\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary, and define $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$. Derive the relations

$$\alpha^*\alpha + \gamma^*\gamma = 1, \qquad \alpha\alpha^* + q^2\gamma\gamma^* = 1, \qquad \gamma^*\gamma = \gamma\gamma^*,$$
$$\alpha\gamma = q\gamma\alpha, \qquad \alpha\gamma^* = q\gamma^*\alpha,$$

and prove that Δ is a coassociative $*$-homomorphism satisfying the cancellation conditions.

Definition 9.8 (Representation-category duality)**.** Let $\mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type}\,1}(U_q(\mathfrak{g}_{\mathbb{C}}))$ be the category of finite-dimensional type-1 unitary modules and intertwiners, and let $\mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q))$ be the category of finite-dimensional unitary corepresentations and intertwiners. For a module (V, π_V), define $U_V \in \mathrm{End}(V) \otimes \mathrm{Pol}(G_q)$ by

$$(\mathrm{id} \otimes \langle x, \cdot \rangle)(U_V) = \pi_V(x) \qquad (x \in U_q(\mathfrak{g}_{\mathbb{C}})).$$

Define

$$\mathcal{F} : \mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type}\,1}(U_q(\mathfrak{g}_{\mathbb{C}})) \longrightarrow \mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q)), \qquad \mathcal{F}(V) = U_V, \quad \mathcal{F}(T) = T,$$

with identity tensor constraints on the underlying spaces $V \otimes W$.

Exercise 9.7 (Matrix coefficients and tensor products)**.** For $v \in V$, $w \in W$, $\varphi \in V^*$, and $\psi \in W^*$, prove

$$c_{\varphi,v}c_{\psi,w} = c_{\varphi\otimes\psi, v\otimes w}.$$

Use the Hopf-pairing identities to prove

$$(\mathrm{id} \otimes \Delta)(U_V) = U_{V,12}U_{V,13}, \qquad U_{V\otimes W} = U_{V,13}U_{W,23}.$$

Prove that $\mathcal{F}$ is fully faithful and essentially surjective and that its tensor constraints are intertwiners. Conclude

$$\mathrm{Rep}^{\mathrm{fd}}_{\mathrm{type}\,1}(U_q(\mathfrak{g}_{\mathbb{C}})) \simeq \mathrm{Corep}^{\mathrm{fd}}(\mathrm{Pol}(G_q))$$

as monoidal $*$-categories.

Definition 9.9 (Conjugates and quantum dimension)**.** For a finite-dimensional H-module V, its *left dual* $V^\vee$ has action

$$(x\varphi)(v) = \varphi(S(x)v),$$

evaluation $\mathrm{ev}_V(\varphi \otimes v) = \varphi(v)$, and coevaluation coev_V. In a rigid C^*-tensor category, a *conjugate* of V is an object $\overline{V}$ with morphisms

$$R_V : \mathbf{1} \longrightarrow \overline{V} \otimes V, \qquad \overline{R}_V : \mathbf{1} \longrightarrow V \otimes \overline{V}$$

satisfying

$$(\overline{R}_V^* \otimes \mathrm{id}_V)(\mathrm{id}_V \otimes R_V) = \mathrm{id}_V, \qquad (R_V^* \otimes \mathrm{id}_{\overline{V}})(\mathrm{id}_{\overline{V}} \otimes \overline{R}_V) = \mathrm{id}_{\overline{V}}.$$

The adjoints R_V^* and $\overline{R}_V^*$ are evaluation maps, and R_V and $\overline{R}_V$ are coevaluation maps. A solution is *standard* when

$$R_V^*(\mathrm{id}_{\overline{V}} \otimes T)R_V = \overline{R}_V^*(T \otimes \mathrm{id}_{\overline{V}})\overline{R}_V \qquad (T \in \mathrm{End}(V)).$$

For a standard solution, define

$$\mathrm{tr}_q(T) = R_V^*(\mathrm{id}_{\overline{V}} \otimes T)R_V, \qquad \dim_q(V) = \mathrm{tr}_q(\mathrm{id}_V).$$

Exercise 9.8 ($SU_q(2)$ fusion and dimensions)**.** Let V_n be the irreducible type-1 $U_q(\mathfrak{sl}_2)$-module of highest weight n. Construct its conjugate equations and prove

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Prove that the normalized positive operators from the Woronowicz orthogonality relations implement the standard solutions and compute

$$\dim_q(V_n) = [n+1]_q = \frac{q^{n+1} - q^{-(n+1)}}{q - q^{-1}}.$$

Prove

$$\dim V_n = n+1, \qquad \lim_{q\to 1} \dim_q(V_n) = n+1, \qquad \dim_q(V_n) > n+1$$

for $0 < q < 1$ and $n \geq 1$.

Definition 9.10 (Tannaka–Krein reconstruction)**.** Let $\mathcal{C}$ be an essentially small rigid C^*-tensor category and let $F : \mathcal{C} \to \mathrm{Hilb}_{\mathrm{fd}}$ be a faithful unitary tensor functor. Define

$$\mathcal{A}_F = \left(\bigoplus_{V \in \operatorname{Ob}\mathcal{C}} F(V)^* \otimes F(V) \right) \Big/ \left\langle (\psi \circ F(T)) \otimes v - \psi \otimes F(T)v \right\rangle,$$

where $T : V \to W$, $v \in F(V)$, and $\psi \in F(W)^*$. Its multiplication is induced by tensor products. With $q_V(\varphi \otimes v) = [\varphi \otimes v]_V$, define

$$\Delta[\varphi \otimes v]_V = (q_V \otimes q_V)(\varphi \otimes \operatorname{coev}_{F(V)}(1) \otimes v), \qquad \varepsilon[\varphi \otimes v]_V = \varphi(v).$$

The involution and antipode are induced by the conjugate data.

Exercise 9.9 (Tannaka–Krein theorem)**.** Prove that the coproduct is well-defined on the quotient and coassociative and that $\mathcal{A}_F$ is a Hopf $*$-algebra. For the forgetful fiber functor

$$F : \operatorname{Rep}(G_q) \longrightarrow \mathrm{Hilb}_{\mathrm{fd}},$$

prove that

$$[\varphi \otimes v]_V \longmapsto c_{\varphi,v}$$

is a Hopf $*$-isomorphism $\mathcal{A}_F \cong \operatorname{Pol}(G_q)$.

Definition 9.11 (Universal R-matrix and braiding)**.** A *universal R-matrix* for a Hopf algebra H is an invertible element $\mathcal{R}$ in a completion of $H \otimes H$ on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x), \quad (\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

For finite-dimensional modules V, W, define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

The pair $(H, \mathcal{R})$ is *quasitriangular.*

Exercise 9.10 (Hexagon identities and the Yang–Baxter equation)**.** Prove that $c_{V,W}$ is a natural module intertwiner and that the two coproduct identities for $\mathcal{R}$ give the two hexagon identities. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on $V^{\otimes 3}$,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators $c_{j,j+1}$ define braid-group representations on tensor powers of every finite-dimensional type-1 module.

Definition 9.12 (Ribbon categories)**.** Let $\mathcal{C}$ be a braided rigid $\mathbb{C}$-linear monoidal category. A *twist* is a natural isomorphism $\theta : \mathrm{id}_{\mathcal{C}} \to \mathrm{id}_{\mathcal{C}}$ satisfying

$$\theta_{\mathbf{1}} = \mathrm{id}_{\mathbf{1}}, \qquad \theta_{X\otimes Y} = c_{Y,X}c_{X,Y}(\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object X.

Exercise 9.11 (Ribbon elements)**.** For a quasitriangular Hopf algebra $(H, \mathcal{R})$, set

$$u = m(S \otimes \mathrm{id})(\mathcal{R}_{21}).$$

Let $v \in H$ be central and invertible and satisfy

$$v^2 = uS(u), \qquad S(v) = v, \qquad \varepsilon(v) = 1, \qquad \Delta(v) = (\mathcal{R}_{21}\mathcal{R})^{-1}(v \otimes v).$$

Prove that

$$\theta_V = \pi_V(v^{-1})$$

defines a ribbon twist on the category of finite-dimensional H-modules. Derive invariance of the quantum trace under framed ribbon isotopy.

Exercise 9.12 (Quantum dimensions and the Chebyshev identity)**.** For the irreducible $SU_q(2)$ objects V_n, use $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$ to prove

$$d_{n+1} = (q + q^{-1})d_n - d_{n-1}, \qquad d_0 = 1, \quad d_1 = q + q^{-1},$$

where $d_n = \dim_q(V_n)$. If U_n is the Chebyshev polynomial determined by $U_0(x) = 1$, $U_1(x) = 2x$, and $U_{n+1}(x) = 2xU_n(x) - U_{n-1}(x)$, deduce

$$\dim_q(V_n) = U_n\left(\frac{q+q^{-1}}{2}\right) = \frac{q^{n+1} - q^{-(n+1)}}{q - q^{-1}}.$$

Exercise 9.13 (q-binomial output statistics)**.** For $0 < q < 1$ and $p > 0$, define

$$Z_n(p,q) = \prod_{j=0}^{n-1}(1 + pq^{2j}), \qquad \mathbb{P}_{n,p,q}(r) = \frac{q^{r(r-1)}\begin{bmatrix} n \\ r \end{bmatrix}_{q^2} p^r}{Z_n(p,q)}.$$

Use the terminating q-binomial identity to prove that this is a probability distribution. Compute its probability-generating function, mean, and variance from $Z_n(sp,q)/Z_n(p,q)$. Derive the distribution by iterating the $U_q(\mathfrak{sl}_2)$ coproduct on a two-channel q-deformed splitter, with r the detected occupation of the first channel. Prove that $q \to 1$ gives the ordinary binomial law with splitting probability $p/(1+p)$. Deduce the predicted deformation of the output count histogram and give a ratio of three adjacent probabilities from which q can be inferred independently of the overall detection rate.

10 Hecke and Double-Affine Spectral Theory

10.1 Hecke algebras and quantum Schur–Weyl duality

Definition 10.1 (Root data and affine Weyl groups)**.** Let E be a finite-dimensional Euclidean space. A finite set $R \subset E \setminus \{0\}$ is a *reduced crystallographic root system* if it is contained in no proper linear subspace of E, is preserved by the reflections

$$s_\alpha(x) = x - \langle x, \alpha^\vee \rangle \alpha, \qquad \alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle},$$

satisfies $R \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$, and has $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in R$. Define

$$W = \langle s_\alpha : \alpha \in R \rangle, \qquad Q^\vee = \langle \alpha^\vee : \alpha \in R \rangle_{\mathbb{Z}}, \qquad W_{\mathrm{aff}} = Q^\vee \rtimes W.$$

The connected components of the complement of the affine reflection hyperplanes

$$H_{\alpha,m} = \{x \in E : \langle x, \alpha \rangle = m\}, \qquad \alpha \in R, \quad m \in \mathbb{Z},$$

are the *alcoves*. After a choice of positive roots, let α_i be the simple roots and $s_i = s_{\alpha_i}$. Then $(W, \{s_i\})$ is a *Coxeter system* with relations $(s_i s_j)^{m_{ij}} = 1$, and $\ell(w)$ is the minimum length of a word in the s_i representing w.

Definition 10.2 (Finite Hecke algebras)**.** Let $(W, \{s_i\})$ be a Coxeter system, and let m_{ij} be the order of $s_i s_j$. Its *Hecke algebra* $\mathcal{H}_t(W)$ over $\mathbb{C}(t^{1/2})$ is generated by T_i with the braid relations

$$\underbrace{T_i T_j T_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{T_j T_i T_j \cdots}_{m_{ij} \text{ factors}}$$

and the quadratic relations

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

Definition 10.3 (Quantum Schur–Weyl actions)**.** Let v be generic, let $U_v(\mathfrak{gl}_N)$ act on its vector module $V = \mathbb{C}^N$, and set $t = v^2$. On $V^{\otimes r}$ define

$$\rho_r(T_i) = \mathrm{id}_V^{\otimes(i-1)} \otimes \check{R}_{V,V} \otimes \mathrm{id}_V^{\otimes(r-i-1)}, \qquad \check{R}_{V,V} = \tau(\pi_V \otimes \pi_V)(\mathcal{R}),$$

with the normalization for which $\check{R}_{V,V}$ has eigenvalues v and $-v^{-1}$.

Exercise 10.1 (Quantum Schur–Weyl duality)**.** Prove that the coproduct action of $U_v(\mathfrak{gl}_N)$ commutes with $\rho_r(T_i)$. Derive the braid and Hecke relations from the universal R-matrix and prove that ρ_r defines an action of $\mathcal{H}_{v^2}(S_r)$. For generic v and $N \geq r$, prove the double-centralizer identities

$$\rho_r(\mathcal{H}_{v^2}(S_r)) = \mathrm{End}_{U_v(\mathfrak{gl}_N)}(V^{\otimes r}), \qquad \pi_r(U_v(\mathfrak{gl}_N)) = \mathrm{End}_{\mathcal{H}_{v^2}(S_r)}(V^{\otimes r}),$$

and the decomposition

$$V^{\otimes r} \cong \bigoplus_{\substack{\lambda \vdash r \\ \ell(\lambda) \leq N}} L_\lambda^{(N)} \otimes S_t^\lambda.$$

Definition 10.4 (Affine Hecke algebras)**.** The extended affine Hecke algebra $\mathcal{H}_t^{\mathrm{aff}}(GL_r)$ is generated by $T_1, \ldots, T_{r-1}$ and commuting invertible elements $X_1, \ldots, X_r$, subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \qquad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of $\mathbb{Z}^r \rtimes S_r$.

Definition 10.5 (Affine quantum Schur–Weyl functor)**.** Let $U_v(L\mathfrak{gl}_N)$ be the quantum loop algebra and let

$$\widehat{V} = V \otimes \mathbb{C}[z^{\pm 1}]$$

be the affinization of its vector evaluation module. The tensor power $\widehat{V}^{\otimes r}$ is a $U_v(L\mathfrak{gl}_N)$–$\mathcal{H}^{\mathrm{aff}}_{v^2}(GL_r)$ bimodule, with the right action determined by the normalized spectral R-matrices and the loop operators. Define

$$\mathsf{SW}^{\mathrm{aff}}_{N,r}(M) = \widehat{V}^{\otimes r} \otimes_{\mathcal{H}^{\mathrm{aff}}_{v^2}(GL_r)} M.$$

Exercise 10.2 (Affine quantum Schur–Weyl duality)**.** Use the spectral Yang–Baxter equation to prove that the two actions on $\widehat{V}^{\otimes r}$ commute and satisfy the affine braid relations. Use the two eigenvalues of the vector R-matrix to derive the affine Hecke quadratic relation. For $N > r$, prove that $\mathsf{SW}^{\mathrm{aff}}_{N,r}$ is an equivalence from finite-dimensional $\mathcal{H}^{\mathrm{aff}}_{v^2}(GL_r)$-modules to the full subcategory of finite-dimensional polynomial $U_v(L\mathfrak{gl}_N)$-modules generated by subquotients of tensor products of r vector evaluation modules.

Exercise 10.3 (Hecke symmetrizers and the Poincaré identity)**.** For $w \in S_r$, let $\ell(w)$ be its Coxeter length. Insert the letter r into a permutation of S_{r-1} and prove

$$\sum_{w \in S_r} t^{\ell(w)} = \prod_{j=1}^{r} \frac{1-t^j}{1-t}.$$

Construct the one-row and one-column idempotents of $\mathcal{H}_t(S_r)$ by weighted sums of the standard Hecke elements indexed by S_r and show that their normalizing constants are the displayed t-factorial and its t^{-1} analogue. Identify their images under quantum Schur–Weyl duality with the quantum symmetric and exterior powers of the vector module.

10.2 DAHA, Askey–Wilson polynomials, and Macdonald theory

10.2.1 Double affine Hecke algebras

Definition 10.6 (Double-affine braid group)**.** Let $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ and let

$$\mathrm{Conf}_r(T^2) = \{(z_1, \dots, z_r) \in (T^2)^r : z_i \neq z_j \text{ for } i \neq j\}.$$

The *elliptic braid group* is

$$B^{\mathrm{ell}}_r = \pi_1(\mathrm{Conf}_r(T^2)/S_r).$$

Its standard generators are adjacent braids T_i and loops X_j, Y_j moving the jth point around the two fundamental cycles of T^2. The central extension in which the total X–Y commutator is denoted q is the *double-affine braid group* of type GL_r.

Exercise 10.4 (From affine to double-affine braids)**.** Derive a presentation of B^{ell}_r from the motions T_i, X_j, Y_j. Prove that the T_i satisfy the finite braid relations, that the X_j commute, that the Y_j commute, and that conjugation by T_i permutes the adjacent loop generators. Compute the central commutator of the two total torus motions. Cutting T^2 along one cycle, prove that the remaining generators form an affine braid group.

Definition 10.7 (Double affine Hecke algebras)**.** Let $B^{\mathrm{da}}(R)$ be the double-affine braid group of a root datum R, with braid generators T_i and central parameter q. For nonzero parameters q and t_i, define

$$\mathsf{H}_{q,\mathbf{t}}(R) = \mathbb{C}(q,\mathbf{t})[B^{\mathrm{da}}(R)] \Big/ \left\langle \mathsf{q} - q, (T_i - t_i^{1/2})(T_i + t_i^{-1/2}) \right\rangle.$$

This is the *double affine Hecke algebra* of R.

Definition 10.8 (Rank-one DAHA and its spherical representation)**.** The universal DAHA $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ is generated by $t_i^{\pm 1}$ for $0 \le i \le 3$, subject to

$$t_i t_i^{-1} = 1, \qquad t_i + t_i^{-1} \text{ central}, \qquad t_0 t_1 t_2 t_3 = q^{-1}.$$

For $k_i \neq 0$, the specialization

$$t_i + t_i^{-1} = k_i + k_i^{-1}$$

is the four-parameter rank-one DAHA. After the standard scalar normalization, let T be its finite Hecke generator with $(T-k)(T+k^{-1}) = 0$, and define

$$e = \frac{T + k^{-1}}{k + k^{-1}}.$$

Its basic polynomial representation is on $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$ by reflection–q-difference operators, and

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}].$$

The action of $e\widehat{\mathsf{H}}_q e$ on $e\mathcal{L}$ is the *spherical rank-one polynomial representation.*

Definition 10.9 (Askey–Wilson operator and polynomials)**.** For $0 < q < 1$ and nonzero a, b, c, d, define

$$A(z) = \frac{(1-az)(1-bz)(1-cz)(1-dz)}{(1-z^2)(1-qz^2)}$$

and, on $f(z) = f(z^{-1})$,

$$(\mathcal{D}_{AW} f)(z) = A(z)\big(f(qz) - f(z)\big) + A(z^{-1})\big(f(q^{-1}z) - f(z)\big).$$

For $x = (z + z^{-1})/2$, the *Askey–Wilson polynomial* is

$$\begin{aligned} p_n(x; a, b, c, d \mid q) &= a^{-n} (ab, ac, ad; q)_n \\ &\qquad \times {}_4\phi_3\left(\begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix} ; q, q \right). \end{aligned}$$

Its weight on the unit circle is

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_\infty}.$$

Exercise 10.5 (Rank-one DAHA and the Askey–Wilson theorem)**.** Construct the basic representation of $\widehat{\mathsf{H}}_q(C_1^\vee, C_1)$ from the involutions

$$(s_1 f)(z) = f(z^{-1}), \qquad (s_0 f)(z) = f(qz^{-1})$$

and first-order reflection–q-difference operators. Verify the four Hecke relations and the product relation. Prove that the spherical subalgebra is generated in its polynomial representation by multiplication by $z + z^{-1}$ and $\mathcal{D}_{AW}$. Prove

$$\mathcal{D}_{AW} p_n = (q^{-n} - 1)(1 - abcdq^{n-1}) p_n.$$

For $a, b, c, d \in (-1, 1)$, prove that $\mathcal{D}_{AW}$ is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z) \overline{g(z)} w_{AW}(z) \frac{dz}{2\pi i z},$$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

10.2.2 Macdonald theory

Definition 10.10 (Type-A polynomial representation)**.** Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}],$$

let X_j act by multiplication by x_j, and let s_i exchange x_i and x_{i+1}. Define

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \qquad 1 \le i < r,$$

and

$$(\pi f)(x_1, \dots, x_r) = f(x_2, \dots, x_r, q x_1).$$

For generic independent parameters q, t, the operator algebra generated by $X_j^{\pm 1}$, T_i, and $\pi^{\pm 1}$ is the polynomial realization of the GL_r *double affine Hecke algebra* $\mathsf{H}_{q,t}(GL_r)$. The subalgebra generated by T_i and $\pi^{\pm 1}$ is an extended affine Hecke algebra; its translation elements form a commuting family $Y_1, \dots, Y_r$ of q-difference-reflection operators.

Exercise 10.6 (Polynomial representation and the DAHA quotient)**.** Prove that the displayed operators preserve $\mathcal{P}_r$ and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \le i < r), \qquad \pi X_r \pi^{-1} = q X_1,$$

and identify the two affine-Hecke subalgebras generated respectively by T_i, X_j and by T_i, Y_j. Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to q.

Definition 10.11 (Macdonald operators and polynomials)**.** Let

$$(T_{q,x_i} f)(x_1, \dots, x_r) = f(x_1, \dots, q x_i, \dots, x_r).$$

The first type-A *Macdonald operator* is

$$D_{q,t} = \sum_{i=1}^{r} \left(\prod_{j \ne i} \frac{t x_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

For generic q, t, the *Macdonald polynomial* $P_\lambda(x; q, t)$ is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^{r} q^{\lambda_i} t^{r-i} \right) P_\lambda, \qquad \lambda_1 \ge \dots \ge \lambda_r \ge 0.$$

For $0 < q, t < 1$, define

$$\Delta_{q,t}(x) = \prod_{i \ne j} \frac{(x_i/x_j; q)_\infty}{(t x_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^{r} \frac{dx_j}{2\pi i x_j}.$$

Exercise 10.7 (DAHA Fourier transform and Macdonald spectral theory)**.** Construct the commuting Y-operators and prove that their S_r-invariant Laurent polynomials preserve $\mathcal{P}_r^{S_r}$. Identify the first elementary symmetric polynomial in the Y_j with $D_{q,t}$ up to normalization. Prove existence, uniqueness, orthogonality, and joint diagonalization of the Macdonald polynomials. Construct the DAHA Fourier transform exchanging the X- and Y-polynomial representations and derive its projective $SL_2(\mathbb{Z})$ relations.

Exercise 10.8 (Ordinary and basic hypergeometric degeneration)**.** Set $q = e^{-\varepsilon}$. Prove

$$\lim_{\varepsilon \to 0^+} \frac{(q^a; q)_m}{(1-q)^m} = (a)_m, \qquad \lim_{q \to 1^-} (1-q)^{1-a} \frac{(q; q)_\infty}{(q^a; q)_\infty} = \Gamma(a).$$

Deduce

$$\lim_{q \to 1^-} {}_r\phi_{r-1}\left(\begin{matrix} q^{a_1}, \ldots, q^{a_r} \\ q^{b_1}, \ldots, q^{b_{r-1}} \end{matrix}; q, z\right) = {}_rF_{r-1}\left(\begin{matrix} a_1, \ldots, a_r \\ b_1, \ldots, b_{r-1} \end{matrix}; z\right).$$

Derive Gauss's summation from the q-Gauss summation. Prove

$$\lim_{q \to 1^-} (1-q)^{-3n} p_n\left(\frac{q^{ix} + q^{-ix}}{2}; q^a, q^b, q^c, q^d \mid q\right) = W_n(x^2; a, b, c, d).$$

Definition 10.12 (Root-system basic hypergeometric weights)**.** Write $f(z^{\pm m}) = f(z^m) f(z^{-m})$ and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^\epsilon z_j^\delta).$$

For $|q|, |t|, |a_s| < 1$, define

$$\Delta_{BC_r}(z) = \prod_{i=1}^r \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1}; q)_\infty} \times \prod_{1 \le i < j \le r} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}.$$

The associated *Gustafson integral* uses the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

Exercise 10.9 (Root systems and Gustafson integrals)**.** Prove that Δ_{BC_r} is invariant under permutations and independent inversions of the z_i. Identify its factors with the roots $\pm e_i$, $\pm 2e_i$, and $\pm e_i \pm e_j$. For $r = 1$, recover the Askey–Wilson weight and its order-two Weyl normalization. Set $t = q^g$ with $g \in \mathbb{N}$, reduce the quotients of q-products to finite products, and compute the limit $q \to 1^-$ with $z_i = e^{i\theta_i}$.

Exercise 10.10 (Askey–Wilson transition spectrum)**.** Use the normalized Askey–Wilson polynomials in $L^2(S^1, w_{AW}(z)\, dz/(2\pi i z))$ as the spectral states of the self-adjoint Hamiltonian $\widehat{H} = \varepsilon \mathcal{D}_{AW}$, where $\varepsilon > 0$ is an energy scale. Prove that its spectral values are

$$E_n = \varepsilon(q^{-n} - 1)(1 - abcdq^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1}) p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to $z + z^{-1}$ produces only neighboring transitions, with resonance frequencies $|E_{n+1} - E_n|/\hbar$ and intensities determined by A_n and C_{n+1}. Show how the measured sequence of line locations determines q and the product $abcd$, while the relative line strengths separate the remaining Askey–Wilson parameters.

11 Quantum Spherical Functions and qKZ Transport

Definition 11.1 (Quantum spherical functions)**.** Let $H = U_q(\mathfrak{g})$ and let $B \subseteq H$ be a right coideal subalgebra, so $\Delta(B) \subseteq B \otimes H$. It plays the role of the stabilizer algebra of a quantum homogeneous space. If an H-module V contains a vector v satisfying

$$bv = \varepsilon(b)v \qquad (b \in B),$$

then the matrix coefficient

$$\varphi_V(x) = \langle v, \pi_V(x)v \rangle$$

is a *quantum zonal spherical function.* It belongs to the coordinate Hopf dual of H and is invariant under the left and right quantum stabilizer actions.

Exercise 11.1 (Flatness, qKZ transport, and affine Hecke monodromy)**.** Let a vector-valued function satisfy the adjacent exchange rule

$$\Psi(\dots, z_i, z_{i+1}, \dots) = \check{R}_i(z_i/z_{i+1})\Psi(\dots, z_{i+1}, z_i, \dots).$$

Show that consistency of the two routes reversing three adjacent variables is exactly the spectral Yang–Baxter equation. Add a periodic rule that returns z_1 at the end as qz_1, and explain why translations enlarge the finite braid action to an affine one. If the R-matrices satisfy a Hecke quadratic, prove that the exchange monodromy factors through an affine Hecke algebra. Write the resulting reflection and translation operators in the same generators used in the polynomial representation defined above. For a unitary specialization of the R-matrix, prepare an incoming three-channel state and prove that the two exchange orders give identical output probabilities. Diagonalize the periodic qKZ transport operator and express its interference fringes through its eigenphases and quantum spherical matrix coefficients. Deduce the predicted equality of the two three-body scattering count distributions and the phase shift produced by one affine transport cycle.

12 Elliptic Hypergeometric Functions and Solvable Lattice Models

Definition 12.1 (Elliptic shifted factorials and gamma function)**.** For $|p|, |q| < 1$, define

$$\theta(z; p) = (z; p)_\infty (p/z; p)_\infty, \qquad (a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

$$(a_1, \dots, a_s; q, p)_m = \prod_{i=1}^{s} (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1}q^{k+1}/z}{1 - p^j q^k z}.$$

Definition 12.2 (Hypergeometric deformation hierarchy)**.** For a series $\sum_m c_m$, set $h_m = c_{m+1}/c_m$. It is *ordinary hypergeometric* when h_m is rational in m, and *basic hypergeometric* when h_m is rational in q^m. It is *elliptic hypergeometric* when, after writing $x = q^m$, its term ratio $h(x)$ is multiplicatively p-periodic:

$$h(px) = h(x).$$

Its parameters are *balanced* when the products of the numerator and denominator parameters required by this ellipticity agree. The deformation and degeneration orders are

$$\text{ordinary/additive} \longrightarrow \text{basic/multiplicative} \longrightarrow \text{elliptic},$$

$$\text{elliptic} \xrightarrow{p\to 0} \text{basic} \xrightarrow{q\to 1} \text{ordinary}.$$

Exercise 12.1 (Elliptic functional equations and degenerations)**.** Prove

$$\theta(pz;p) = -z^{-1}\theta(z;p), \qquad \Gamma(qz;p,q) = \theta(z;p)\Gamma(z;p,q),$$

$$\Gamma(pz;p,q) = \theta(z;q)\Gamma(z;p,q), \qquad \Gamma(z;p,q)\Gamma(pq/z;p,q) = 1.$$

Prove

$$\theta(z;0) = 1 - z, \qquad (a;q,0)_m = (a;q)_m, \qquad \Gamma(z;0,q) = \frac{1}{(z;q)_\infty}.$$

Definition 12.3 (Frenkel–Turaev series and elliptic beta integral)**.** For $a^2q^{n+1} = bcde$, define

$$\begin{aligned} {}_{10}V_9(a;b,c,d,e,q^{-n};q,p) = \sum_{m=0}^{n} &\frac{\theta(aq^{2m};p)}{\theta(a;p)} \\ &\times \frac{(a,b,c,d,e,q^{-n};q,p)_m}{(q,aq/b,aq/c,aq/d,aq/e,aq^{n+1};q,p)_m} q^m. \end{aligned}$$

For parameters $t_1,\dots,t_6$ satisfying

$$|t_j| < 1, \qquad \prod_{j=1}^{6} t_j = pq,$$

the *elliptic beta integral* is

$$\frac{(p;p)_\infty (q;q)_\infty}{4\pi i} \int_{|z|=1} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)} \frac{dz}{z}.$$

Exercise 12.2 (Frenkel–Turaev and elliptic beta evaluations)**.** Prove the Frenkel–Turaev summation

$${}_{10}V_9(a;b,c,d,e,q^{-n};q,p) = \frac{(aq,aq/(bc),aq/(bd),aq/(cd);q,p)_n}{(aq/b,aq/c,aq/d,aq/(bcd);q,p)_n}.$$

Prove the elliptic beta evaluation

$$\frac{(p;p)_\infty (q;q)_\infty}{4\pi i} \int_{|z|=1} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)} \frac{dz}{z} = \prod_{1\le j<k\le 6} \Gamma(t_j t_k;p,q).$$

Take $p \to 0$ in both identities and identify the resulting terminating very-well-poised ${}_8\phi_7$ summation and basic hypergeometric beta integral. Obtain the Askey–Wilson integral by specialization.

Definition 12.4 (Elliptic Ruijsenaars operator)**.** For type A_{r-1}, define

$$D^{\mathrm{ell}}_{p,q,t} = \sum_{i=1}^{r} \left(\prod_{j \neq i} \frac{\theta(tx_i/x_j;p)}{\theta(x_i/x_j;p)} \right) T_{q,x_i}.$$

Exercise 12.3 (Differential, basic, and elliptic operator hierarchy)**.** Construct the commuting higher elliptic Ruijsenaars operators. Prove

$$D^{\mathrm{ell}}_{p,q,t} \longrightarrow D_{q,t} \qquad (p \to 0).$$

Set $q = e^{\varepsilon}$ and $t = e^{g\varepsilon}$ and derive the Heckman–Opdam differential operator from the second-order term as $\varepsilon \to 0$. Take the rational scaling limit of its hyperbolic coefficients and obtain the rational Calogero–Moser operator. Organize the eigenfunction limits as

$$\text{elliptic Ruijsenaars functions} \longrightarrow \text{Macdonald polynomials} \longrightarrow \text{Heckman–Opdam functions.}$$

Exercise 12.4 (The Calogero–Sutherland limit)**.** Write $x_i = e^{z_i}$, set $q = e^{\epsilon}$ and $t = e^{g\epsilon}$, and expand $D_{q,t}$ through order ϵ^2. After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i<j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put $z_i = i\theta_i$ and conjugate by

$$\Psi_0(\theta) = \prod_{i<j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g .$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = -\sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for $r = 2$, degree two. Identify their Sutherland eigenvalues.

Exercise 12.5 (Selberg measure and the Jacobi beta ensemble)**.** For coupling $g > 0$, set $\beta = 2g$. For $a, b > 0$, take the BC_N Calogero–Sutherland ground state in Jacobi coordinates $x_i \in (0,1)$,

$$\Psi_0(x) = C \prod_{i=1}^{N} x_i^{(a-1)/2} (1 - x_i)^{(b-1)/2} \prod_{i<j} |x_i - x_j|^{\beta/2}.$$

Prove

$$|\Psi_0(x)|^2 = \frac{1}{Z^{\mathrm{Jac}}_{N,\beta}(a,b)} \prod_{i=1}^{N} x_i^{a-1} (1 - x_i)^{b-1} \prod_{i<j} |x_i - x_j|^{\beta},$$

where the Selberg integral gives

$$Z^{\mathrm{Jac}}_{N,\beta}(a,b) = \prod_{j=0}^{N-1} \frac{\Gamma(a + j\beta/2)\Gamma(b + j\beta/2)\Gamma(1 + (j+1)\beta/2)}{\Gamma(a + b + (N + j - 1)\beta/2)\Gamma(1 + \beta/2)}.$$

Interpret $|\Psi_0|^2 dx_1 \cdots dx_N$ as the joint law of N random eigenvalues. For the type-A circular ground state, prove instead

$$|\Psi_0(\theta)|^2 \propto \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^{\beta}.$$

12.1 Elliptic star–triangle and Yang–Baxter identities

Definition 12.5 (Star–triangle data)**.** A family of edge weights $W_u(x,y)$, one-spin weights $S(x)$, and scalar normalizations $R(u,v,w)$ is *star–triangle data* when

$$\int S(x)W_u(a,x)W_v(b,x)W_w(c,x)\,dx = R(u,v,w)W_{u'}(b,c)W_{v'}(a,c)W_{w'}(a,b)$$

for its parameter constraint and contour. It is *elliptic* when its weights are finite products and ratios of $\Gamma(cdot\,; p,q)$.

Exercise 12.6 (Elliptic beta integral and star–triangle relation)**.** Parametrize the six variables in the elliptic beta integral by three spectral parameters and three external spins. Derive an elliptic star–triangle identity from its evaluation. Apply the identity twice to a three-row lattice strip and prove equality of the two local reorderings that yields commuting transfer matrices. Take $p \to 0$ and then $q \to 1$ and derive the corresponding trigonometric and rational star–triangle relations. Compare this integral construction with the $6j$-symbol/recoupling construction of the Yang–Baxter equation. For fixed boundary spins, prove that the normalized boundary-spin probabilities are unchanged by the local star–triangle replacement. Construct the commuting row transfer matrices, let $\Lambda_0(u)$ and $\Lambda_1(u)$ be their two largest eigenvalues in modulus, and derive the correlation-length prediction

$$\xi(u)^{-1} = -\log|\Lambda_1(u)/\Lambda_0(u)|.$$

State how equality of partition-function ratios, local spin frequencies, and the extracted correlation length tests the elliptic identity, and how the $p \to 0$ and $q \to 1$ limits change these observables.

13 Root-of-Unity Degeneration and Tensor Categories

Definition 13.1 (Root-of-unity semisimplification)**.** Fix $k \in \mathbb{N}$ and

$$q = e^{\pi i/(k+2)}.$$

Let $\mathcal{T}_k$ be the ribbon category of finite-dimensional type-1 tilting modules for the Lusztig integral form of $U_q(\mathfrak{sl}_2)$ specialized at q. A morphism $f : X \to Y$ is *negligible* when

$$\mathrm{tr}_q(gf) = 0 \qquad \text{for every } g : Y \to X.$$

Let $\mathcal{N}_k$ be the negligible morphisms and define

$$\mathcal{C}_k = \mathrm{Kar}(\mathcal{T}_k/\mathcal{N}_k),$$

where Kar denotes idempotent completion.

Exercise 13.1 (Semisimplified $SU(2)_k$ fusion)**.** Prove that $\mathcal{N}_k$ is a tensor ideal preserved by duals, braiding, and twist. Prove that $\mathcal{C}_k$ is semisimple with simple objects $V_0, \ldots, V_k$ and that

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b|\le c\le \min(a+b,2k-a-b)\\ a+b+c\equiv 0 \pmod 2}} V_c.$$

Compute

$$\dim_q(V_a) = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

Definition 13.2 (Modular tensor categories)**.** A *fusion category* is a semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit, finite-dimensional morphism spaces, and finitely many simple-object classes. A *modular tensor category* is a ribbon fusion category for which the matrix

$$\widetilde{S}_{ij} = \operatorname{tr}_q(c_{X_j,X_i} c_{X_i,X_j})$$

on the simple objects $(X_i)_{i\in I}$ is invertible. Define

$$d_i = \dim_q(X_i), \qquad \mathcal{D} = \left(\sum_{i\in I} d_i^2\right)^{1/2}, \qquad S = \mathcal{D}^{-1}\widetilde{S}, \qquad T_{ij} = \delta_{ij}\theta_i,$$

where $\theta_{X_i} = \theta_i \operatorname{id}_{X_i}$. It is *unitary* when it is a C^*-tensor category and its braiding and twist are unitary.

Exercise 13.2 (Modularity of $SU(2)_k$)**.** Prove that $\mathcal{C}_k$ is a unitary modular tensor category. For $0 \le a, b \le k$, compute

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \qquad \theta_a = \exp\left(\frac{\pi i\, a(a+2)}{2(k+2)}\right).$$

Prove directly that S is unitary and invertible. For simple charges a, b, prove that the normalized monodromy amplitude measured by winding a once around b is

$$M_{ab} = \frac{S_{ab}S_{00}}{S_{0a}S_{0b}}.$$

In a two-path anyon interferometer with path intensities I_1, I_2, derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2}\,\Re(e^{i\phi} M_{ab}).$$

Insert the sine formula for S_{ab} and predict the fringe visibility and phase for every enclosed charge b. Identify pairs of charges that the chosen probe a cannot distinguish and prove that the full family of probe charges distinguishes them because S is invertible.

Part III

Counting

14 Exact State Counting and Modular Spectroscopy

14.1 Partitions and generating functions

14.1.1 Partition generating functions and finite boxes

Definition 14.1 (Partitions and q-Pochhammer symbols)**.** An *integer partition* of $n \ge 0$ is a finite nonincreasing sequence $\lambda = (\lambda_1, \dots, \lambda_r)$ of positive integers with $|\lambda| = \sum_i \lambda_i = n$. Let $p(n)$ be the number of partitions, with $p(0) = 1$. For a formal variable q, set

$$(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty}(1 - aq^j).$$

Analytically, the infinite product converges for $|q| < 1$. Euler's partition generating function is

$$P(q) = \sum_{n\geq 0} p(n)q^n = \frac{1}{(q;q)_\infty}.$$

Exercise 14.1 (Bosonic and fermionic occupation numbers)**.** Consider oscillator modes with one-particle energies $\hbar\omega, 2\hbar\omega, 3\hbar\omega, \ldots$. Show that a bosonic Fock state of excitation energy $n\hbar\omega$ is a partition of n, and derive

$$\prod_{m\geq 1} \frac{1}{1-q^m} = \sum_{n\geq 0} p(n)q^n.$$

For fermionic occupation numbers in $\{0,1\}$, show that $\prod_{m\geq 1}(1+q^m)$ counts partitions into distinct parts. Construct a bijection, by separating powers of two in the multiplicities, between partitions into distinct parts and partitions into odd parts, and conclude

$$\prod_{m\geq 1}(1+q^m) = \prod_{m\geq 1} \frac{1}{1-q^{2m-1}}.$$

Interpret the two products as different quasiparticle descriptions with the same exact degeneracies.

Definition 14.2 (Gaussian binomial coefficients)**.** For $0 \leq r \leq n$, the *Gaussian binomial coefficient* is

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \frac{(q;q)_n}{(q;q)_r (q;q)_{n-r}},$$

and it is zero outside this range.

Exercise 14.2 (Gaussian binomial theorem)**.** Derive the recurrence

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \begin{bmatrix} n-1 \\ r \end{bmatrix}_q + q^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_q.$$

Prove that $\left[\begin{smallmatrix} n \\ r \end{smallmatrix}\right]_q$ is a polynomial in q with nonnegative integer coefficients and that the coefficient of q^m counts partitions of m whose Young diagrams fit inside an $r \times (n-r)$ rectangle.

14.1.2 Rogers–Ramanujan identities and exclusion rules

Definition 14.3 (Rogers–Ramanujan sum and product series)**.** For $|q| < 1$, define

$$G_{\text{sum}}(q) = \sum_{r\geq 0} \frac{q^{r^2}}{(q;q)_r}, \qquad H_{\text{sum}}(q) = \sum_{r\geq 0} \frac{q^{r(r+1)}}{(q;q)_r},$$

$$G_{\text{prod}}(q) = \frac{1}{(q;q^5)_\infty (q^4;q^5)_\infty}, \qquad H_{\text{prod}}(q) = \frac{1}{(q^2;q^5)_\infty (q^3;q^5)_\infty}.$$

Exercise 14.3 (Difference conditions and Rogers–Ramanujan counting)**.** Fix the number r of parts. Subtract the staircase $(2r-1, 2r-3, \ldots, 1)$ to prove that $q^{r^2}/(q;q)_r$ generates partitions with exactly r parts whose successive parts differ by at least two. Use the staircase $(2r, 2r-2, \ldots, 2)$ for the same difference condition with smallest part at least two. Interpret the product sides as partitions into parts congruent to 1 or 4 modulo 5, and to 2 or 3 modulo 5, respectively. Compute all four series through q^{30} using finite products and finite sums, and verify coefficientwise that $G_{\text{sum}} = G_{\text{prod}}$ and $H_{\text{sum}} = H_{\text{prod}}$ to that order. Translate the coefficient equalities into finite tests of the two partition identities and identify which further step would be required for a proof in all degrees.

14.1.3 Modular asymptotics

Exercise 14.4 (Partition asymptotics from modular inversion)**.** Put $q = e^{-t}$ and use the modular transformation of η from Exercise 2.11 to prove

$$P(e^{-t}) \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \qquad (t \downarrow 0).$$

Apply a saddle-point argument, or an appropriate Tauberian theorem with its hypotheses stated, to derive the Hardy–Ramanujan formula

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Interpret $\log p(n)$ as the microcanonical entropy of the bosonic modes in the first exercise of this part. Distinguish this asymptotic result from an exact coefficient formula; see Hardy–Ramanujan, *Proceedings of the London Mathematical Society* **17** (1918).

14.2 Characters and exact state counting

14.2.1 Virasoro representations and character formulas

Definition 14.4 (The Virasoro algebra and its characters)**.** The *Virasoro algebra* has generators L_n for $n \in \mathbb{Z}$ and a central element acting by c, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

A highest-weight vector $|h\rangle$ satisfies $L_0|h\rangle = h|h\rangle$ and $L_n|h\rangle = 0$ for $n > 0$. For a positive-energy module H_a with finite-dimensional L_0 eigenspaces, its *character* is

$$\chi_a(\tau) = \mathrm{Tr}_{H_a}\, q^{L_0 - c/24}.$$

The shift by $-c/24$ is the Casimir energy produced by mapping the plane to the cylinder.

Exercise 14.5 (Descendants, partitions, and null states)**.** Use the Virasoro commutation relations to move positive modes to the right and apply the Poincaré–Birkhoff–Witt theorem to deduce the Verma-module character

$$\chi^{\mathrm{Verma}}_{c,h}(\tau) = \frac{q^{h-c/24}}{(q;q)_\infty}.$$

For the vacuum, impose $L_{-1}|0\rangle = 0$ and find the resulting generic vacuum descendant product. Explain why additional null vectors subtract states and turn unrestricted partition numbers into the more structured q-series of minimal models.

14.2.2 Modular invariance and exact state counting

Definition 14.5 (Rational characters and modular invariants)**.** A finite family of positive-energy sectors H_a is a *rational modular character system* when its character vector carries a representation of the modular group,

$$\chi_a(-1/\tau) = \sum_b S_{ab}\chi_b(\tau), \qquad \chi_a(\tau+1) = \sum_b T_{ab}\chi_b(\tau).$$

A full torus partition function has the form

$$Z(\tau,\bar\tau)=\sum_{a,b}M_{ab}\chi_a(\tau)\overline{\chi_b(\tau)}$$

with nonnegative integer multiplicities. It is modular invariant when M commutes with S and T.

Exercise 14.6 (Exact state counting in the Ising CFT)**.** For the Ising theory with $c=1/2$ and primary weights $0,1/2,1/16$, take

$$\begin{aligned}\chi_0&=\frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}}+\sqrt{\frac{\vartheta_4}{\eta}}\right),\\ \chi_{1/2}&=\frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}}-\sqrt{\frac{\vartheta_4}{\eta}}\right),\\ \chi_{1/16}&=\frac{1}{\sqrt{2}}\sqrt{\frac{\vartheta_2}{\eta}}.\end{aligned}$$

Expand each as $q^{h-c/24}$ times a power series and interpret its first coefficients as exact descendant degeneracies. Derive

$$S=\frac{1}{2}\begin{pmatrix}1&1&\sqrt{2}\\1&1&-\sqrt{2}\\\sqrt{2}&-\sqrt{2}&0\end{pmatrix},\qquad T_{aa}=e^{2\pi i(h_a-c/24)}.$$

Verify the modular invariance of $|\chi_0|^2+|\chi_{1/2}|^2+|\chi_{1/16}|^2$. Using the Verlinde formula

$$N_{ab}{}^{c}=\sum_x\frac{S_{ax}S_{bx}\overline{S_{cx}}}{S_{0x}},$$

recover the Ising fusion rules. Relate these towers to the critical Ising spectroscopy exercise in Quantum Topology.

Exercise 14.7 (The Cardy density of states)**.** Let a modular-invariant unitary CFT have discrete spectrum, unique vacuum, and left and right central charges $c,\bar c$. Set $\tau=i\beta/(2\pi)$ and use S-invariance to show that the high-temperature partition function is controlled by the vacuum at inverse temperature $4\pi^2/\beta$. A saddle-point inverse Laplace transform then gives, for large left and right excitation numbers,

$$\log\rho(N,\bar N)\sim 2\pi\sqrt{\frac{cN}{6}}+2\pi\sqrt{\frac{\bar c\bar N}{6}}.$$

State how c is replaced by $c_{\mathrm{eff}}=c-24h_{\min}$ in a nonunitary theory. For a nonrotating state with $c=\bar c$ and $N=\bar N$, recover the thermal entropy used in the BTZ exercise and hence reproduce the horizon entropy from boundary-state counting; see Cardy, *Nuclear Physics B* **270** (1986).

14.2.3 Integrable lattice models

Definition 14.6 (Transfer matrices and integrability)**.** For a two-dimensional lattice model on a cylinder, a *row transfer matrix* $T_N(u)$ propagates one width-N row to the next and depends on a spectral parameter u. The model is *integrable* when it admits a nontrivial commuting family

$$[T_N(u),T_N(v)]=0\qquad\text{for every } u,v,$$

usually obtained from a Yang–Baxter or star–triangle relation. Commuting transfer matrices permit simultaneous spectral analysis, but integrability alone does not guarantee that every finite-size degeneracy has a simple closed formula.

Exercise 14.8 (Hard hexagons and exact finite-volume counting)**.** In the hard-hexagon model, occupied vertices of a triangular lattice may not be adjacent. On an $N \times M$ torus define

$$Z_{N,M}(z) = \sum_{I \text{ independent}} z^{|I|}.$$

Index a row transfer matrix by cyclic binary rows and set its matrix element to zero when two consecutive rows violate nearest-neighbor exclusion, and otherwise to $z^{(|a|+|b|)/2}$. Prove $Z_{N,M}(z) = \mathrm{Tr}(T_N(z)^M)$ and compute the polynomial for several small sizes. Show that the coefficients are exact state counts, whereas the largest eigenvalue controls the thermodynamic free energy.

Verify the algebraic identity

$$z_c = \frac{11 + 5\sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2}\right)^5,$$

and, for the finite matrices already constructed, evaluate the largest eigenvalue and its first two z-derivatives near z_c. Plot their change with N and explain why finite matrices remain analytic even when their large-size limit may develop a singularity.

Exercise 14.9 (Theta-character predictions for finite-size spectra)**.** For a critical periodic system of circumference L and propagation speed v, take

$$H_L = E_{\mathrm{bulk}} L + \frac{2\pi v}{L}\left(L_0 + \overline{L}_0 - \frac{c+\bar{c}}{24}\right), \qquad P_L = \frac{2\pi}{L}(L_0 - \overline{L}_0).$$

Insert the Ising characters expressed through $\vartheta_2, \vartheta_3, \vartheta_4$, and η into the modular-invariant partition function. Prove that the coefficient of $q^{h+N-c/24}\bar{q}^{\bar{h}+\bar{N}-\bar{c}/24}$ is the predicted multiplicity at

$$E - E_{\mathrm{bulk}} L = \frac{2\pi v}{L}\left(h + \bar{h} + N + \bar{N} - \frac{c+\bar{c}}{24}\right), \quad P = \frac{2\pi}{L}(h - \bar{h} + N - \bar{N}).$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the L^{-1} Casimir-energy correction, and state how finite-size energy levels or transfer-matrix eigenvalues determine c, the primary weights, and v.

15 Probability, Random Matrices, and Statistical Universality

15.1 Probability spaces and convergence

Definition 15.1 (Probability spaces and convergence)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map $X : \Omega \to Y$, its *law* is $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B))$, and, when it is scalar-valued and integrable, its *expectation* is its integral. A random vector $X = (X_1, \ldots, X_N)$ has *joint density* p with respect to a measure μ when

$$\mathbb{P}(X \in B) = \int_B p \, d\mu.$$

The Gaussian law $N(m, v)$ on $\mathbb{R}$ has density

$$\frac{1}{\sqrt{2\pi v}} \exp\left[-\frac{(x-m)^2}{2v}\right].$$

A sequence of random variables in a metric space (Y, d) *converges in probability* to X when

$$\mathbb{P}(d(X_N, X) > \varepsilon) \longrightarrow 0 \qquad (\varepsilon > 0).$$

It *converges in distribution*, written $X_N \Longrightarrow X$, when

$$\int f \, d\mathbb{P}_{X_N} \longrightarrow \int f \, d\mathbb{P}_X$$

for every bounded continuous f.

Exercise 15.1 (Gaussian moments and Hermite orthogonality)**.** Let $X \sim N(0,1)$. Complete the square and use the Gaussian integral to prove

$$\mathbb{E}[e^{tX}] = e^{t^2/2}.$$

Define the probabilists' Hermite polynomials by

$$e^{xt - t^2/2} = \sum_{n=0}^{\infty} \mathrm{He}_n(x) \frac{t^n}{n!}.$$

Using two copies of the generating function, prove

$$\mathbb{E}[\mathrm{He}_m(X) \, \mathrm{He}_n(X)] = n! \, \delta_{mn}.$$

Relate them to the oscillator polynomials by $\mathrm{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2})$, and derive

$$\mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}, \qquad \mathbb{E}[X^{2n+1}] = 0.$$

15.2 Random matrices and statistical universality

Definition 15.2 (Vandermonde determinant and Dyson classes)**.** For $\lambda = (\lambda_1, \ldots, \lambda_N)$, define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^{N}.$$

Let $\mathbb{F} = \mathbb{R}, \mathbb{C}$, or $\mathbb{H}$ and set

$$\beta = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}.$$

The Gaussian orthogonal, unitary, and symplectic ensembles, denoted GOE, GUE, and GSE, consist of the self-adjoint matrices $H = H^*$ over $\mathbb{R}, \mathbb{C}$, and $\mathbb{H}$, respectively, with density

$$d\mathbb{P}_{N,\beta}(H) = C_{N,\beta} \exp\left[-\frac{\beta N}{4\sigma^2} \operatorname{Tr}(H^2)\right] dH.$$

Here H^* is conjugate transpose, and dH is the Euclidean volume induced by the invariant Hilbert–Schmidt inner product $\langle X, Y \rangle_{\mathrm{HS}} = \operatorname{Re} \operatorname{Tr}(XY)$ on the real space of self-adjoint matrices.

Exercise 15.2 (Eigenvalue–orbit Jacobian)**.** Use the exterior-power determinant formula to prove

$$\Delta(\lambda) = \prod_{i<j} (\lambda_j - \lambda_i).$$

For each Dyson class, let G_β act on the self-adjoint matrices by conjugation and let T_β be the stabilizer of a regular diagonal matrix. Compute the Jacobian of the conjugation-orbit map

$$(G_\beta/T_\beta) \times \mathbb{R}^N_{\mathrm{reg}} \longrightarrow \mathrm{Herm}_N(\mathbb{F}), \qquad (UT_\beta, \lambda) \longmapsto U\,\mathrm{diag}(\lambda)U^{-1},$$

with respect to the invariant Hilbert–Schmidt metric, and prove

$$dH = C\,|\Delta(\lambda)|^\beta \prod_{i=1}^N d\lambda_i\, d\mu_{G_\beta/T_\beta},$$

with the stabilizer and permutation factors included in C. Integrate over the conjugation orbits and derive

$$p_{N,\beta}(\lambda) = \frac{1}{Z_{N,\beta}} \exp\left[-\frac{\beta N}{4\sigma^2}\sum_{i=1}^N \lambda_i^2\right] \prod_{i<j} |\lambda_i - \lambda_j|^\beta.$$

Definition 15.3 (Beta ensembles and Coulomb gases)**.** For an interval $I \subseteq \mathbb{R}$, a confining potential V, and $\beta > 0$, the *beta ensemble* is

$$d\mathbb{P}_{N,\beta,V}(\lambda) = \frac{1}{Z_{N,\beta,V}} \prod_{i<j} |\lambda_i - \lambda_j|^\beta \prod_{i=1}^N e^{-\beta N V(\lambda_i)/2}\, d\lambda_i.$$

Its *Coulomb-gas Hamiltonian* is

$$\mathcal{H}_N(\lambda) = \frac{N}{2}\sum_{i=1}^N V(\lambda_i) - \sum_{i<j} \log|\lambda_i - \lambda_j|, \qquad d\mathbb{P}_{N,\beta,V} = Z^{-1} e^{-\beta \mathcal{H}_N}\, d\lambda.$$

The Jacobi beta ensemble is the Selberg law on $[0,1]^N$, and the circular beta ensemble is

$$\frac{1}{Z^{\mathrm{circ}}_{N,\beta}} \prod_{i<j} |e^{i\theta_i} - e^{i\theta_j}|^\beta \prod_{i=1}^N \frac{d\theta_i}{2\pi}.$$

Exercise 15.3 (Logarithmic equilibrium and the semicircle law)**.** For the empirical measure $L_N = N^{-1}\sum_i \delta_{\lambda_i}$, derive the mean-field energy

$$\mathcal{I}_V(\mu) = \int V(x)\, d\mu(x) - \iint \log|x-y|\, d\mu(x)d\mu(y).$$

Prove that its minimizer satisfies

$$V(x) - 2\int \log|x-y|\, d\mu_V(y) = \ell$$

on its support. For $V(x) = x^2/(2\sigma^2)$, solve the singular integral equation and obtain

$$\rho_{\mathrm{sc}}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2 - x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x).$$

Definition 15.4 (Wigner matrices and empirical spectral measures)**.** A *Wigner matrix* H_N is an $N \times N$ Hermitian random matrix whose upper-triangular entries are independent and centered, with $\mathbb{E}|H_{ij}|^2 = \sigma^2/N$ off the diagonal and, for every fixed m,

$$\sup_{N,i,j} \mathbb{E}|\sqrt{N} H_{ij}|^m < \infty.$$

Its *empirical spectral measure* is

$$L_{H_N} = \frac{1}{N}\sum_{j=1}^N \delta_{\lambda_j(H_N)}, \qquad \int x^k\, dL_{H_N}(x) = \frac{1}{N}\,\mathrm{Tr}(H_N^k).$$

Exercise 15.4 (Wigner semicircle law)**.** Expand $N^{-1}\mathbb{E}\,\mathrm{Tr}(H_N^k)$ as a sum over closed index walks. Prove that only noncrossing pair-matched walks survive as $N \to \infty$ and that their number is the Catalan number. Prove that the variance tends to zero and conclude that L_{H_N} converges in probability, in moments, to $\rho_{\mathrm{sc}}(x)dx$.

Definition 15.5 (Unfolded spectra and level repulsion)**.** For a smooth approximation $\overline{N}(E)$ to the cumulative level count, define

$$x_j = \overline{N}(E_j), \qquad s_j = x_{j+1} - x_j.$$

The variables x_j are the *unfolded levels.* Their local mean spacing is one. Exact symmetry sectors are unfolded separately. The factor $|\Delta(\lambda)|^\beta$ defines *level repulsion* of order β.

Exercise 15.5 (Wigner surmises and repulsion exponents)**.** For 2×2 GOE and GUE matrices, separate the trace from the traceless coordinates and compute the radial distribution of the eigenvalue gap. After rescaling to unit mean spacing, derive

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \qquad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Use the beta-ensemble density to prove

$$P_\beta(s) \sim c_\beta s^\beta \qquad (s \downarrow 0),$$

and compare it with the Poisson spacing law e^{-s}.

Definition 15.6 (Global, bulk, and edge scales)**.** Order the eigenvalues as $\lambda_1 \leq \cdots \leq \lambda_N$. Define the classical location γ_j by

$$\frac{j}{N} = \int_{-2\sigma}^{\gamma_j} \rho_{\mathrm{sc}}(x)\, dx.$$

At $E \in (-2\sigma, 2\sigma)$, the bulk coordinate is

$$u = N\rho_{\mathrm{sc}}(E)(\lambda - E),$$

and at the upper soft edge it is

$$\xi = \frac{N^{2/3}}{\sigma}(\lambda - 2\sigma).$$

Exercise 15.6 (Bulk and edge scales)**.** Use the expected count $N\rho_{\mathrm{sc}}(E)dE$ to derive the bulk spacing scale $[N\rho_{\mathrm{sc}}(E)]^{-1}$. Prove

$$\rho_{\mathrm{sc}}(2\sigma - s) \sim \frac{1}{\pi\sigma^{3/2}}\sqrt{s} \qquad (s \downarrow 0),$$

and derive the edge scale $s \asymp \sigma N^{-2/3}$.

Definition 15.7 (Airy kernel and the Tracy–Widom law)**.** Let Ai be the solution of $y'' = xy$ decaying at $+\infty$. Define

$$K_{\mathrm{Ai}}(x,y) = \frac{\mathrm{Ai}(x)\,\mathrm{Ai}'(y) - \mathrm{Ai}'(x)\,\mathrm{Ai}(y)}{x - y}$$

by continuity on the diagonal. The GUE Tracy–Widom distribution is

$$F_2(s) = \det(I - K_{\mathrm{Ai}})_{L^2(s,\infty)}.$$

15.2.1 Correlation functions and point processes

Definition 15.8 (Point processes and correlation functions)**.** Let (X, μ) be a locally compact measure space. A *simple point process* is a random locally finite counting measure

$$\Xi = \sum_i \delta_{x_i}$$

with distinct points almost surely. Its k-point correlation function ρ_k is defined by

$$\mathbb{E} \sum_{i_1,\ldots,i_k}^{\neq} f(x_{i_1}, \ldots, x_{i_k}) = \int_{X^k} f(x_1, \ldots, x_k)\rho_k(x_1, \ldots, x_k) \prod_{j=1}^{k} d\mu(x_j).$$

For a symmetric joint density p_N of exactly N points,

$$\rho_k(x_1, \ldots, x_k) = \frac{N!}{(N-k)!} \int_{X^{N-k}} p_N(x_1, \ldots, x_N) \prod_{j=k+1}^{N} d\mu(x_j).$$

Exercise 15.7 (Factorial moments and gap probabilities)**.** For a relatively compact measurable set B and $N_B = \Xi(B)$, prove, whenever either side is finite,

$$\mathbb{E}(N_B)_k = \int_{B^k} \rho_k \, d\mu^{\otimes k}, \qquad (N_B)_k = N_B(N_B - 1) \cdots (N_B - k + 1).$$

If $\mathbb{E}[2^{N_B}] < \infty$, use inclusion–exclusion and justify the exchange of expectation and summation to prove

$$\mathbb{P}(N_B = 0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{B^k} \rho_k \, d\mu^{\otimes k}.$$

15.2.2 Determinantal/Pfaffian processes and Fredholm determinants

Definition 15.9 (Determinantal and Pfaffian point processes)**.** For an antisymmetric matrix $A = (a_{ij}) \in M_{2m}(\mathbb{C})$, define

$$\mathrm{Pf}(A) = \frac{1}{2^m m!} \sum_{\sigma \in S_{2m}} \mathrm{sgn}(\sigma) \prod_{j=1}^{m} a_{\sigma(2j-1),\sigma(2j)}.$$

A point process is *determinantal* with kernel K when

$$\rho_k(x_1, \ldots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^{k}.$$

It is *Pfaffian* with antisymmetric 2×2 matrix kernel $\mathbb{K}$ when

$$\rho_k(x_1, \ldots, x_k) = \mathrm{Pf}[\mathbb{K}(x_i, x_j)]_{i,j=1}^{k}.$$

For a trace-class integral operator K_B on $L^2(B, \mu)$, its *Fredholm determinant* is

$$\det(I + zK_B) = \sum_{k=0}^{\infty} \frac{z^k}{k!} \int_{B^k} \det[K(x_i, x_j)]_{i,j=1}^{k} \, d\mu^{\otimes k}.$$

The Fredholm Pfaffian is defined by the analogous series with Pfaffians of the matrix kernel. A *positive contraction* is an operator K satisfying $0 \leq K \leq I$.

Exercise 15.8 (Counting statistics and Fredholm formulas)**.** For a determinantal process, prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z-1)K_B), \qquad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Derive the corresponding Fredholm-Pfaffian formula for a Pfaffian process. If K_B is a positive trace-class contraction with eigenvalues (λ_j), prove

$$\det(I + zK_B) = \prod_j (1 + z\lambda_j)$$

and that N_B is distributed as a sum of independent Bernoulli variables with parameters λ_j.

15.2.3 Orthogonal-polynomial ensembles

Definition 15.10 (Orthogonal-polynomial ensembles)**.** Let $w(x)d\mu(x)$ be a measure with finite moments and let p_j be its monic orthogonal polynomials,

$$\int p_j(x)p_k(x)w(x)d\mu(x) = h_j\delta_{jk}.$$

The associated *orthogonal-polynomial ensemble* has density

$$p_N(x_1,\ldots,x_N) = \frac{1}{Z_N}\Delta(x)^2\prod_{i=1}^{N} w(x_i), \qquad Z_N = N!\prod_{j=0}^{N-1} h_j.$$

Its correlation kernel is

$$K_N(x,y) = \sqrt{w(x)w(y)}\sum_{j=0}^{N-1}\frac{p_j(x)p_j(y)}{h_j}.$$

Exercise 15.9 (Andréief identity and determinantal correlations)**.** Prove Andréief's identity

$$\int_{X^N} \det[f_i(x_j)]\det[g_i(x_j)]\prod_{j=1}^{N} d\mu(x_j) = N!\det\left[\int_X f_i(x)g_j(x)d\mu(x)\right].$$

Use the Vandermonde determinant to prove

$$p_N(x_1,\ldots,x_N) = \frac{1}{N!}\det[K_N(x_i,x_j)]_{i,j=1}^{N}$$

and

$$\rho_k(x_1,\ldots,x_k) = \det[K_N(x_i,x_j)]_{i,j=1}^{k}.$$

Exercise 15.10 (Skew-orthogonal ensembles)**.** For an antisymmetric bilinear form $\langle\cdot,\cdot\rangle_s$, construct skew-orthogonal polynomials satisfying

$$\langle p_{2j}, p_{2k+1}\rangle_s = r_j\delta_{jk}, \qquad \langle p_{2j}, p_{2k}\rangle_s = \langle p_{2j+1}, p_{2k+1}\rangle_s = 0.$$

Prove de Bruijn's Pfaffian integration identity and use it to derive the matrix correlation kernels for the GOE and GSE.

15.2.4 Christoffel–Darboux and integrable kernels

Definition 15.11 (Christoffel–Darboux and integrable kernels)**.** Let $q_j = p_j/\sqrt{h_j}$ and let κ_j be the leading coefficient of q_j. The Christoffel–Darboux kernel is

$$K_N(x,y) = \sqrt{w(x)w(y)}\,\frac{\kappa_{N-1}}{\kappa_N}\,\frac{q_N(x)q_{N-1}(y) - q_{N-1}(x)q_N(y)}{x-y}.$$

An *integrable kernel* is a kernel of the form

$$K(x,y) = \frac{f(x)^{\mathsf{T}}g(y)}{x-y}, \qquad f(x)^{\mathsf{T}}g(x) = 0.$$

Exercise 15.11 (Projection and resolvent identities)**.** Derive the Christoffel–Darboux formula from the three-term recurrence. Prove

$$\int K_N(x,z)K_N(z,y)d\mu(z) = K_N(x,y), \qquad \int K_N(x,x)d\mu(x) = N.$$

For an integrable trace-class kernel K, prove that the resolvent $R = K(I-K)^{-1}$ is again integrable and express its numerator using $(I-K)^{-1}f$ and $(I-K^{\mathsf{T}})^{-1}g$.

15.2.5 Bulk, soft-edge, and hard-edge universality

Definition 15.12 (Universal limiting kernels)**.** The bulk, soft-edge, and hard-edge kernels are respectively

$$K_{\sin}(u,v) = \frac{\sin \pi(u-v)}{\pi(u-v)},$$

$$K_{\mathrm{Ai}}(u,v) = \frac{\mathrm{Ai}(u)\,\mathrm{Ai}'(v) - \mathrm{Ai}'(u)\,\mathrm{Ai}(v)}{u-v},$$

and, for $\alpha > -1$,

$$K_{\mathrm{Bes}}^{(\alpha)}(u,v) = \frac{J_\alpha(\sqrt{u})\sqrt{v}\,J'_\alpha(\sqrt{v}) - \sqrt{u}\,J'_\alpha(\sqrt{u})J_\alpha(\sqrt{v})}{2(u-v)}.$$

The values on the diagonal are defined by continuity.

Exercise 15.12 (Classical kernel universality)**.** Use the Christoffel–Darboux formula and the asymptotics of the Hermite, Laguerre, and Jacobi polynomials to prove convergence of their rescaled kernels, where applicable, to $K_{\sin}$ in the bulk, K_{Ai} at a soft edge, and $K_{\mathrm{Bes}}^{(\alpha)}$ at a hard edge. Deduce convergence of every correlation function and of the associated Fredholm gap probabilities. Recover the Tracy–Widom law for the largest GUE eigenvalue. For an unfolded spectrum, derive from $K_{\sin}$ the nearest-neighbor level-repulsion exponent and number variance, and state the corresponding predictions for a chaotic quantum spectrum in the unitary symmetry class. From K_{Ai} derive the $N^{-2/3}$ edge-fluctuation scale and the limiting distribution of the largest resonance or covariance eigenvalue. Specify the unfolding and centering required before comparing either prediction with spectral data.

16 Integrable Probability and KPZ Universality

16.1 RSK and last-passage percolation

Definition 16.1 (Partitions, tableaux, and Schur functions)**.** A partition is a sequence $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ with finite sum $|\lambda| = \sum_i \lambda_i$. A semistandard Young tableau of shape λ has weakly increasing rows and strictly increasing columns. For variables $a = (a_1, \ldots, a_m)$, define

$$s_\lambda(a) = \sum_T \prod_{i=1}^m a_i^{m_i(T)},$$

where $m_i(T)$ is the number of entries equal to i.

Definition 16.2 (RSK correspondence and last-passage time)**.** The Robinson–Schensted–Knuth construction associates

$$A = (a_{ij}) \longmapsto (P, Q)$$

to a matrix of nonnegative integers a pair of semistandard Young tableaux of the same shape λ, with the row and column sums of A giving the contents of P and Q. For weights $a_{ij} \geq 0$, define

$$G(m, n) = \max_{\pi:(1,1)\nearrow(m,n)} \sum_{(i,j)\in\pi} a_{ij}.$$

Exercise 16.1 (Schensted and Greene theorems)**.** Prove the RSK bijection and Greene's theorem. For $A \in M_{m,n}(\mathbb{Z}_{\geq 0})$, deduce

$$\lambda_1 = G(m, n)$$

and identify $\lambda_1 + \cdots + \lambda_r$ with the maximal weight of r disjoint up-right paths. For independent geometric weights with parameters $a_i b_j$, use RSK to derive

$$\mathbb{P}(\lambda) = \prod_{i,j}(1 - a_i b_j) s_\lambda(a) s_\lambda(b)$$

and prove the Cauchy identity

$$\sum_\lambda s_\lambda(a) s_\lambda(b) = \prod_{i,j}(1 - a_i b_j)^{-1}.$$

16.2 Schur and Macdonald processes

Definition 16.3 (Schur measures and processes)**.** For $\mu \subseteq \lambda$, the *skew Schur function* $s_{\lambda/\mu}$ is the tableau sum over the skew diagram $\lambda \setminus \mu$. A *specialization* is an algebra homomorphism from the algebra of symmetric functions to $\mathbb{C}$; it is *Schur-positive* when $s_{\lambda/\mu}(\rho) \geq 0$ for all $\mu \subseteq \lambda$. The *Schur measure* associated to two Schur-positive specializations ρ^+, ρ^- is

$$\mathbb{P}(\lambda) = \frac{s_\lambda(\rho^+) s_\lambda(\rho^-)}{\sum_\nu s_\nu(\rho^+) s_\nu(\rho^-)}.$$

An ascending *Schur process* is a probability measure on $\lambda^{(1)} \subseteq \cdots \subseteq \lambda^{(m)}$ proportional to

$$s_{\lambda^{(1)}}(\rho_1^+) \prod_{r=2}^m s_{\lambda^{(r)}/\lambda^{(r-1)}}(\rho_r^+) s_{\lambda^{(m)}}(\rho^-).$$

Exercise 16.2 (Schur processes as determinantal processes)**.** Encode a partition by the Maya diagram

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \tfrac{1}{2} : i \geq 1\}.$$

Use the Cauchy and skew-Cauchy identities to prove that the induced space–time point process is determinantal. Derive its double-contour correlation kernel and express the distribution of λ_1 as a Fredholm determinant. Recover geometric last-passage percolation as a one-time marginal.

Definition 16.4 (Macdonald measures and processes)**.** Let $P_\lambda(\,\cdot\,; q, t)$ be the Macdonald polynomials and let $Q_\lambda = b_\lambda(q, t)P_\lambda$ be their dual family. Define the skew functions by

$$P_\lambda(x, y) = \sum_\mu P_{\lambda/\mu}(x)P_\mu(y), \qquad Q_\lambda(x, y) = \sum_\mu Q_{\lambda/\mu}(x)Q_\mu(y).$$

For finite specializations a, b, define

$$\Pi(a; b) = \prod_{i,j} \frac{(ta_ib_j; q)_\infty}{(a_ib_j; q)_\infty}.$$

The *Macdonald measure* is

$$\mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a; q, t)Q_\lambda(b; q, t)}{\Pi(a; b)}.$$

The ascending *Macdonald process* is

$$\frac{P_{\lambda^{(1)}}(a^{(1)}) \prod_{r=2}^m P_{\lambda^{(r)}/\lambda^{(r-1)}}(a^{(r)})Q_{\lambda^{(m)}}(b)}{\Pi(a^{(1)}, \ldots, a^{(m)}; b)}.$$

Exercise 16.3 (Macdonald observables and degenerations)**.** Apply the commuting Macdonald difference operators to the Cauchy identity and derive contour-integral formulas for the joint moments of

$$\sum_i q^{\lambda_i} t^{N-i}.$$

Prove that $q = t$ gives the Schur process, $q = 0$ gives the Hall–Littlewood process, and $t = 0$ gives the q-Whittaker process. Obtain Jack processes from the limit $q = t^\alpha$, $t \uparrow 1$.

16.3 Dyson Brownian motion and KPZ-type dynamics

Definition 16.5 (Brownian motion and space–time white noise)**.** A *standard Brownian motion* is a continuous real process $(B_t)_{t\geq 0}$ with $B_0 = 0$ and independent increments satisfying

$$B_t - B_s \sim N(0, t - s) \qquad (0 \leq s < t).$$

Itô differentials satisfy

$$dB_i\, dB_j = \delta_{ij}\, dt, \qquad dB_i\, dt = dt^2 = 0.$$

Space–time white noise on $\mathbb{R}_+ \times \mathbb{R}$ is the centered Gaussian generalized process ξ satisfying

$$\mathbb{E}[\xi(\varphi)\xi(\psi)] = \int_{\mathbb{R}_+\times\mathbb{R}} \varphi(t, x)\psi(t, x)\, dt\, dx$$

for test functions φ, ψ.

Definition 16.6 (Dyson Brownian motion)**.** For $\beta \geq 1$ and a confining potential V, *Dyson Brownian motion* is the interacting diffusion

$$d\lambda_i = \sqrt{\frac{2}{\beta N}}\, dB_i + \left(\frac{1}{N}\sum_{j\neq i}\frac{1}{\lambda_i - \lambda_j} - \frac{1}{2}V'(\lambda_i)\right) dt.$$

Its reversible density is

$$Z^{-1}\prod_{i<j}|\lambda_i - \lambda_j|^\beta \prod_i e^{-\beta N V(\lambda_i)/2}.$$

Exercise 16.4 (Matrix diffusion and eigenvalue dynamics)**.** Apply Itô calculus to Hermitian and real-symmetric Ornstein–Uhlenbeck matrix processes and derive Dyson Brownian motion for $\beta = 2$ and $\beta = 1$. Prove noncollision, compute the generator in divergence form, and verify the stated reversible measure. Derive the complex Burgers equation for the large-N Stieltjes transform and, in the unitary case, the sine- and Airy-kernel local equilibria.

Definition 16.7 (TASEP and the KPZ equation)**.** In the totally asymmetric simple exclusion process, particles on $\mathbb{Z}$ attempt rate-one jumps from x to $x+1$, and a jump is suppressed when $x+1$ is occupied. Its integrated current defines a height function. The one-dimensional KPZ equation is

$$\partial_t h = \nu\, \partial_x^2 h + \frac{\lambda}{2}(\partial_x h)^2 + \sqrt{D}\, \xi,$$

where ξ is space–time white noise. Its Cole–Hopf solution is $h = (2\nu/\lambda)\log Z$, where Z solves the corresponding stochastic heat equation with multiplicative noise.

Exercise 16.5 (Last passage, particle dynamics, and KPZ scaling)**.** Use the recursion

$$G(m,n) = \max\{G(m-1,n), G(m,n-1)\} + a_{mn}$$

to identify exponential last-passage times with TASEP currents. For i.i.d. exponential weights, prove

$$\frac{G(n,n) - 4n}{2^{4/3}n^{1/3}} \Longrightarrow \chi_2, \qquad \mathbb{P}(\chi_2 \leq s) = F_2(s).$$

Derive the 1:2:3 KPZ fluctuation scaling and its spatial Airy process limit.

Exercise 16.6 (Macdonald dynamics and KPZ degenerations)**.** At $t = 0$, intertwine the q-Whittaker difference operators with the q-TASEP generator. Derive a Fredholm-determinant formula for the q-Laplace transform of a particle position. Take the $q \uparrow 1$ limit to the O'Connell–Yor semi-discrete polymer and then its intermediate-disorder limit to the stochastic heat equation and the KPZ equation. Identify the corresponding degenerations of the Macdonald-process observables. Under narrow-wedge initial data, carry the steepest-descent analysis of the limiting Fredholm kernel to the Airy kernel. After fixing the nonuniversal velocity and scale from the first two cumulants, prove that the centered height has $t^{1/3}$ fluctuations and converges to the GUE Tracy–Widom distribution. Deduce the $t^{2/3}$ correlation-length scale and state the resulting data collapse for repeated interface-height or directed-polymer free-energy measurements.

17 Free Probability and Asymptotic Freeness

17.1 Free independence and additive convolution

Definition 17.1 (Noncommutative probability spaces)**.** A *noncommutative probability space* is a unital complex algebra A with a unital linear functional $\varphi : A \to \mathbb{C}$. Its elements are *random* *variables* and their joint moments are $\varphi(a_1 \cdots a_n)$. In the $*$-algebra setting, a *state* is a positive unital functional; it is *tracial* when $\varphi(ab) = \varphi(ba)$.

Definition 17.2 (Free independence and free cumulants)**.** Unital subalgebras $(A_i)_i$ of (A, φ) are *freely independent* if

$$\varphi(a_1 \cdots a_m) = 0$$

whenever $a_j \in A_{i_j}$, $\varphi(a_j) = 0$, and neighboring indices are distinct. Variables are free when their generated unital subalgebras are.

Let $NC(n)$ be the lattice of noncrossing partitions of $\{1, \dots, n\}$. The *free cumulants* κ_n are the multilinear functionals determined recursively by

$$\varphi(a_1 \cdots a_n) = \sum_{\pi \in NC(n)} \kappa_\pi(a_1, \dots, a_n),$$

where κ_π is the product of the cumulants associated to the blocks of π. Freeness is equivalently the vanishing of every mixed free cumulant.

Definition 17.3 (Semicircular variables)**.** A self-adjoint variable S is *semicircular* with variance σ^2 if its distribution is

$$d\mu_{\mathrm{sc},\sigma}(x) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2}\, \mathbf{1}_{[-2\sigma, 2\sigma]}(x)\, dx.$$

Equivalently, its only nonzero free cumulant is $\kappa_2(S, S) = \sigma^2$.

Exercise 17.1 (Free central limit theorem)**.** Use noncrossing pairings to prove

$$\varphi(S^{2m+1}) = 0, \qquad \varphi(S^{2m}) = C_m \sigma^{2m}, \qquad C_m = \frac{1}{m+1}\binom{2m}{m}.$$

Let $X_1, X_2, \dots$ be freely independent, identically distributed, centered self-adjoint variables with $\varphi(X_i^2) = \sigma^2$ and moments of every order. Use multilinearity and vanishing mixed cumulants to prove that the moments of $n^{-1/2}(X_1 + \cdots + X_n)$ converge to these semicircular moments. Compare this free central limit theorem with the Gaussian limit for classically independent variables.

Definition 17.4 (Cauchy and R-transforms)**.** For a compactly supported probability measure μ on $\mathbb{R}$, its *Cauchy transform* is

$$G_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(t)}{z - t}, \qquad z \in \mathbb{C} \setminus \operatorname{supp} \mu.$$

Near infinity it has a local compositional inverse $K_\mu(w) = G_\mu^{\langle -1 \rangle}(w)$ near $w = 0$. The *R-transform* is defined by

$$K_\mu(w) = \frac{1}{w} + R_\mu(w), \qquad R_\mu(w) = \sum_{n \geq 1} \kappa_n(\mu) w^{n-1}.$$

If freely independent self-adjoint variables X, Y have laws μ, ν, then the law of $X + Y$ is their *free additive convolution* $\mu \boxplus \nu$, characterized by

$$R_{\mu \boxplus \nu} = R_\mu + R_\nu.$$

Exercise 17.2 (Free convolution and the semicircle equation)**.** Derive the additivity of the R-transform from the vanishing of mixed free cumulants. Show that a centered semicircular law of variance σ^2 has $R(w) = \sigma^2 w$ and hence

$$\sigma^2 G(z)^2 - zG(z) + 1 = 0.$$

Select the solution satisfying $zG(z) \to 1$ at infinity and recover the semicircle density from the Stieltjes inversion formula. Deduce that the free sum of semicircular variables of variances σ_1^2 and σ_2^2 is semicircular of variance $\sigma_1^2 + \sigma_2^2$, and rederive the free central limit theorem at the level of transforms.

17.2 Random matrices and asymptotic freeness

Definition 17.5 (Asymptotic freeness)**.** Random matrix families $A_N^{(1)}, \dots, A_N^{(r)}$ are *asymptotically free* when their normalized expected joint trace moments converge to the moments of freely independent variables:

$$\lim_{N\to\infty} \frac{1}{N} \mathbb{E}\,\mathrm{Tr}\, P(A_N^{(1)}, \dots, A_N^{(r)}) = \varphi(P(a_1, \dots, a_r))$$

for every noncommutative polynomial P.

Exercise 17.3 (Asymptotic freeness of independent GUE matrices)**.** For independent GUE matrices $H_N^{(1)}, \dots, H_N^{(r)}$, apply Wick's rule to normalized mixed traces of centered polynomials. Prove that crossing pairings are lower order and that mixed free cumulants vanish in the limit. Conclude that the family converges in joint moments to freely independent semicircular variables.

Exercise 17.4 (Catalan generating function and the semicircle transform)**.** Let $C_n = (n+1)^{-1}\binom{2n}{n}$. Decompose a noncrossing pairing at the partner of its first point to prove

$$C_{n+1} = \sum_{j=0}^{n} C_j C_{n-j}.$$

Deduce

$$\sum_{n=0}^{\infty} C_n t^n = \frac{1 - \sqrt{1-4t}}{2t}.$$

Use the semicircular moment formula to recover

$$G_{\mathrm{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}, \qquad zG_{\mathrm{sc}}(z) \longrightarrow 1 \quad (z \to \infty),$$

and identify the branch cut with the limiting spectral interval of GUE. Use Stieltjes inversion to recover the semicircle density from the jump of G_{sc} across the branch cut. For a measured large Hamiltonian or coupled-mode spectrum modeled by an additive free sum, use $R_{\mu\boxplus\nu} = R_\mu + R_\nu$ to predict the support edges and density of states. State how convergence of the empirical Stieltjes transform and the observed spectral edges test the asymptotic-freeness model.